
Chapter 5

Current Mirrors and Bias Techniques

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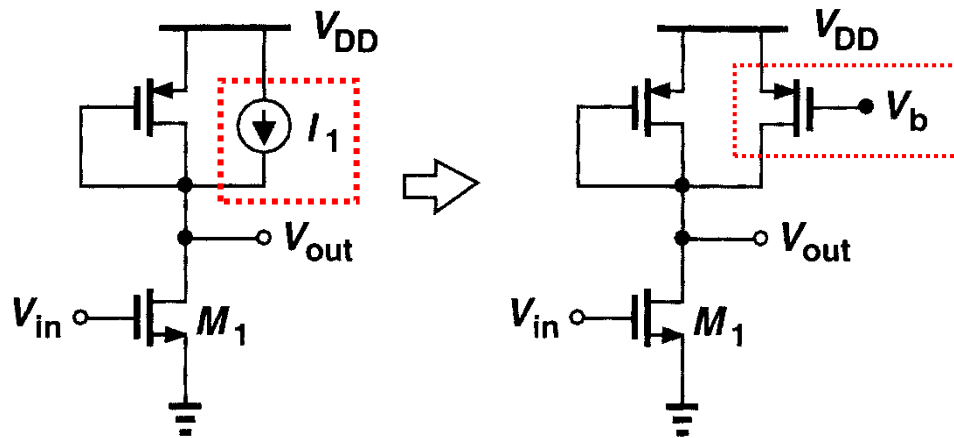
Outline

- **Basic current mirrors**
- Cascode current mirrors
- Active current mirrors
- Bias techniques

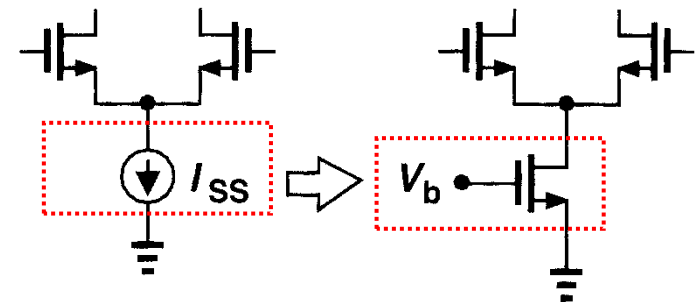


Application of current sources

- An important advantages of CMOS
 - Providing high-performance current sources
- Current sources are basic blocks of analog IC



Load

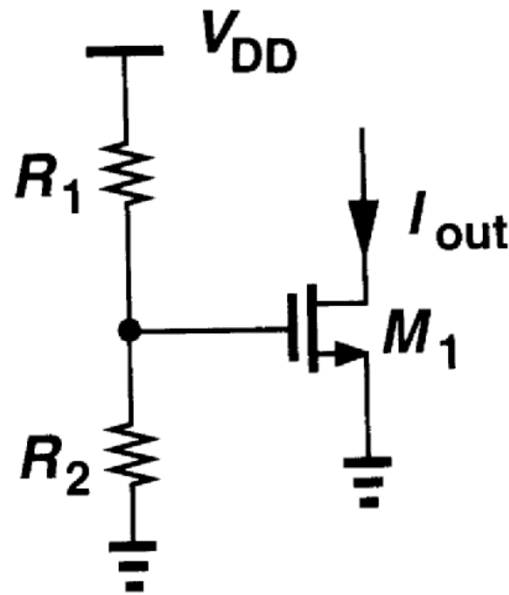


Bias



How to obtain stable current source

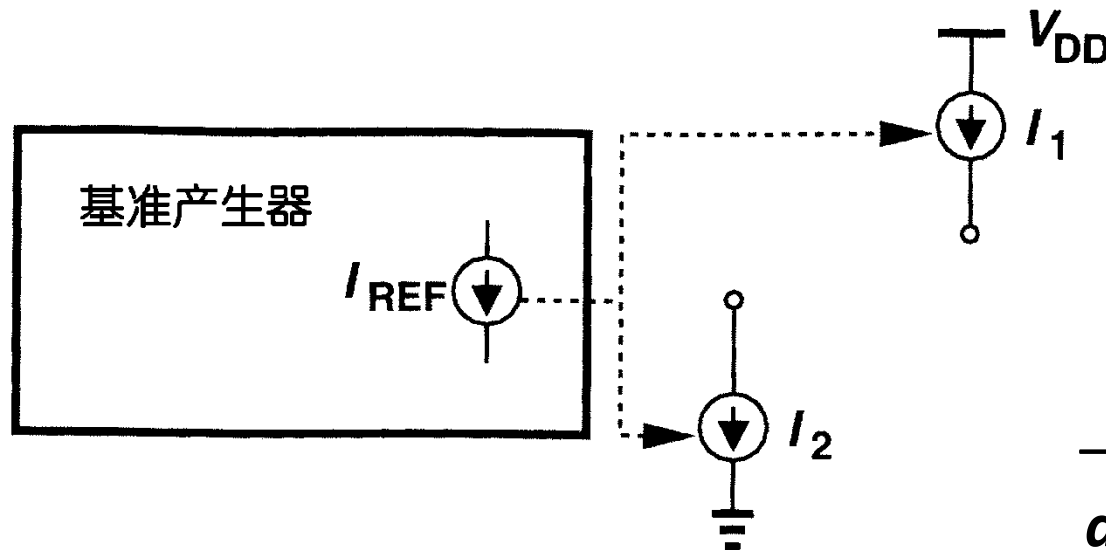
- Simple but poor way: resistive divider.
 - I_{out} is influenced by PVT variations
 - **P**: V_{TH} varies by 100mV from wafer to wafer
 - **V**: V_{DD} varies by $\pm 10\%$
 - **T**: μ_n and V_{TH} are functions of temperature
 - V_{OD} varies from 200mV to 250mV
 - Error in I_{out} can exceed 40%!



$$I_{out} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \left(\frac{R_2}{R_2 + R_1} V_{DD} - V_{TH} \right)^2$$

“Copy” currents from a reference

- Even if the gate voltage is well defined, the current is not
- Solution is to “copy” currents from a well defined current reference
 - How to generate I_{REF} ?---first the egg or the hen?



–Reference generator is discussed in Chapter 12

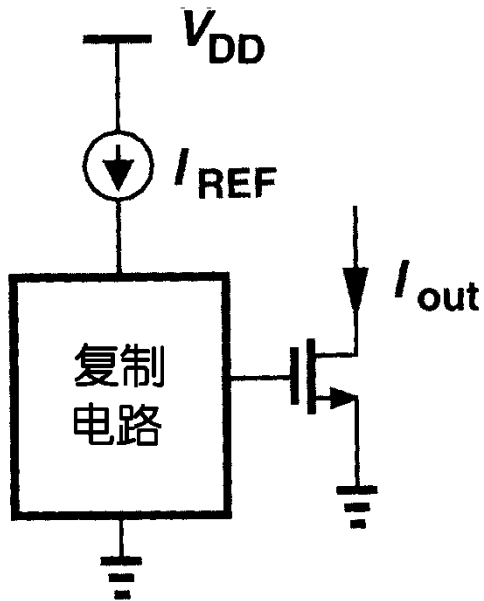


Copy circuit

- How do we generate copies of I_{REF} ?

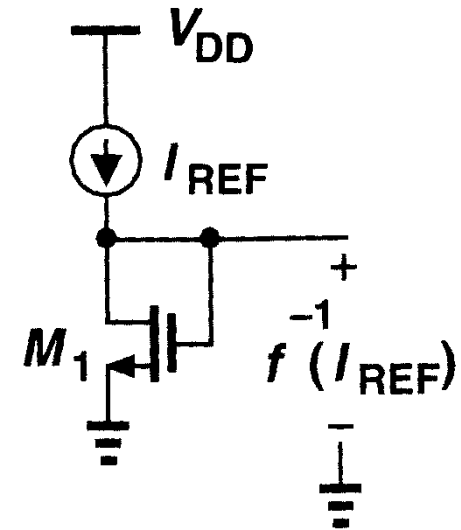
- How to guarantee $I_{out} = I_{REF}$?

- For a MOSFET



$$I_{out} = f(V_{GS})$$

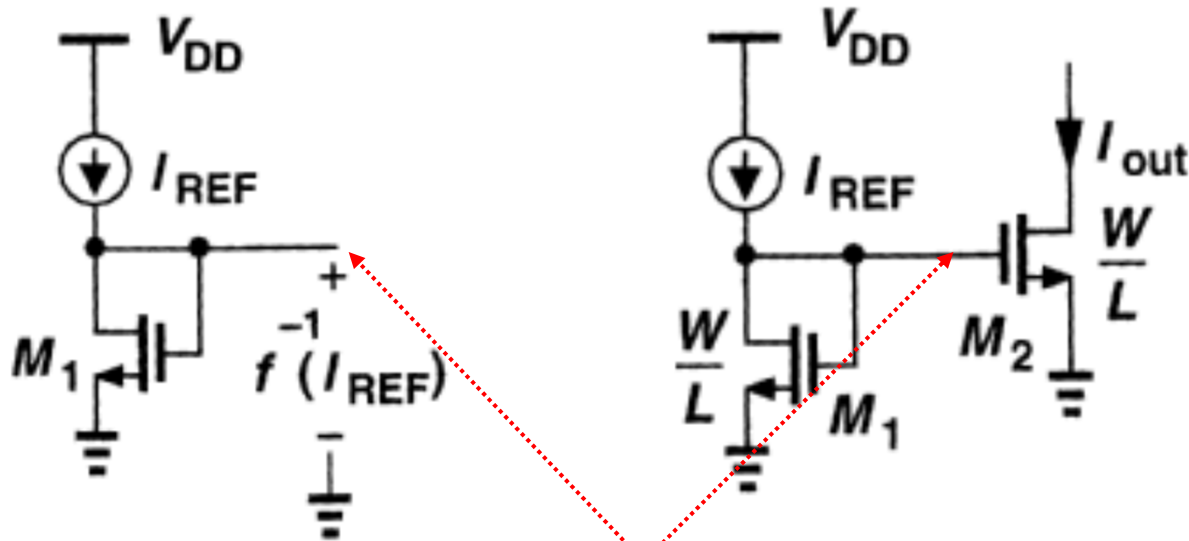
$$V_{GS} = f^{-1}(I_{REF})$$



- Use a diode-connected MOS to generate the inverse function
- Apply this voltage to a second MOS



Basic current mirror



$$V_{GS} = f^{-1}(I_{REF})$$

$$\text{if } (W/L)_1 = (W/L)_2$$

$$\begin{aligned} I_{out} &= f(V_{GS}) \\ &= f f^{-1}(I_{REF}) \\ &= I_{REF} \end{aligned}$$



Use current mirror to generate $n \cdot I_{REF}$

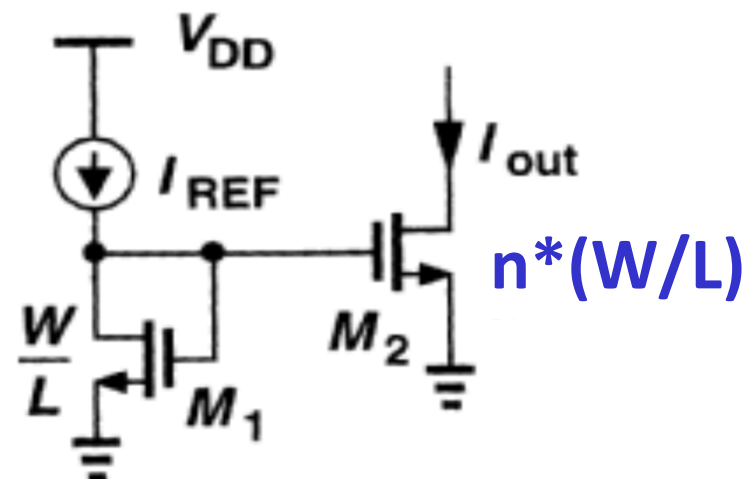
- **If** $(W/L)_2 = n \cdot (W/L)_1$

– Assuming $\lambda = 0$

$$I_{REF} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_1 (V_{GS} - V_{TH})^2$$

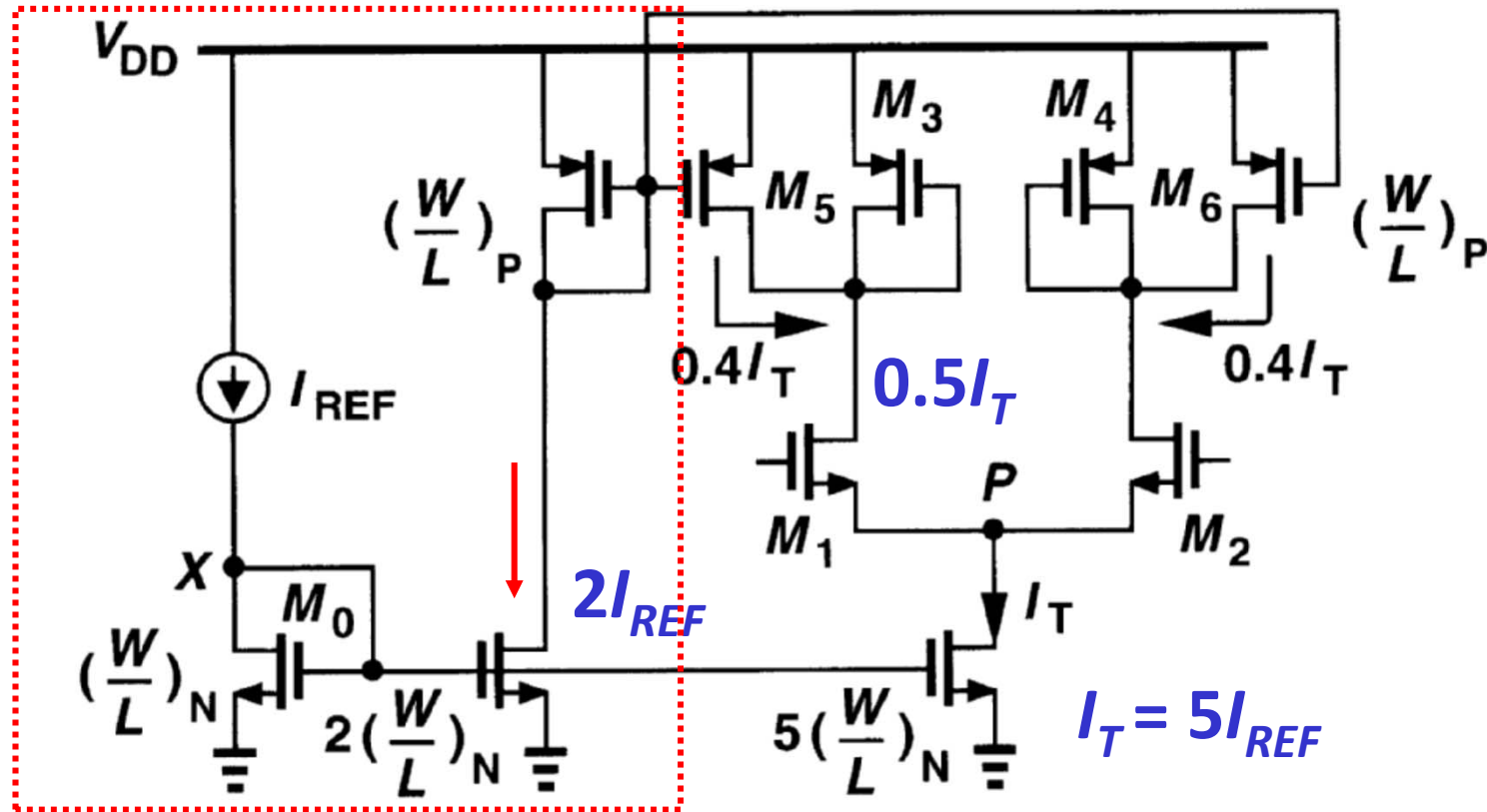
$$I_{out} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_2 (V_{GS} - V_{TH})^2$$

$$I_{out} = \frac{(W/L)_2}{(W/L)_1} I_{REF} = n \cdot I_{REF}$$



- **The ratio is determined by ratio of device dimensions**
 - **Can be controlled with reasonable accuracy**

Amplifier Bias Example



All NMOS (PMOS) have the same L_N (L_P)
Change W_N (W_P) to change the ratio

Example 5.2: current mirror as a current amplifier

- Calculate small-signal gain

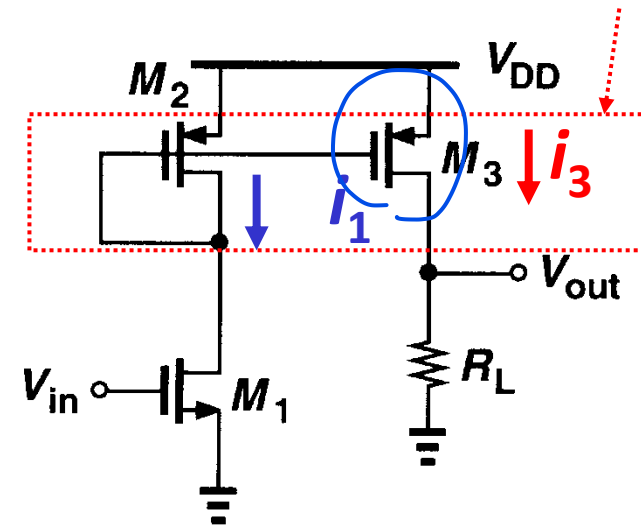
$$i_1 = g_m v_{in}$$

$$i_3 = \frac{(W/L)_3}{(W/L)_1} i_1$$

$$v_{out} = i_3 \cdot R_L$$

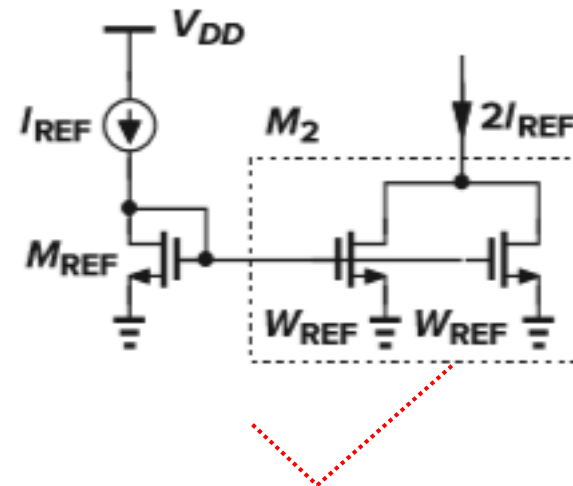
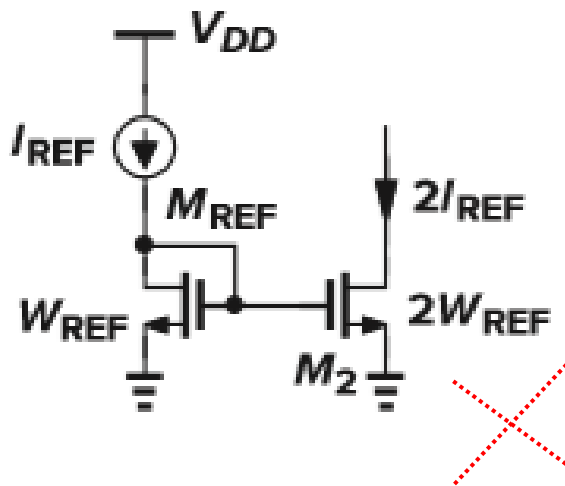
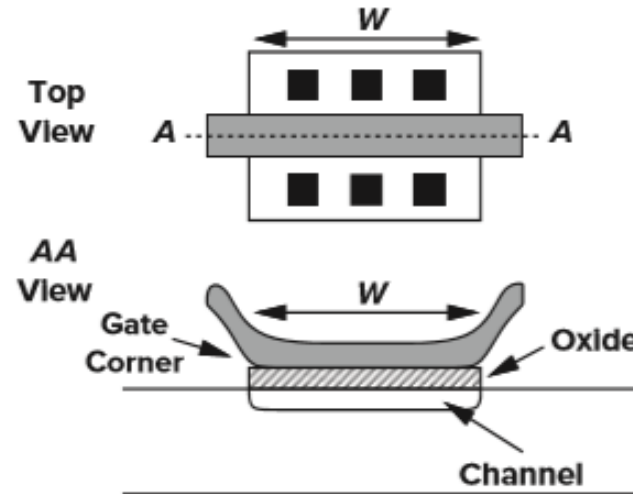
$$\longrightarrow A_v = \frac{v_{out}}{v_{in}} = g_{m1} \cdot \frac{(W/L)_3}{(W/L)_1} \cdot R_L$$

Current amplifier



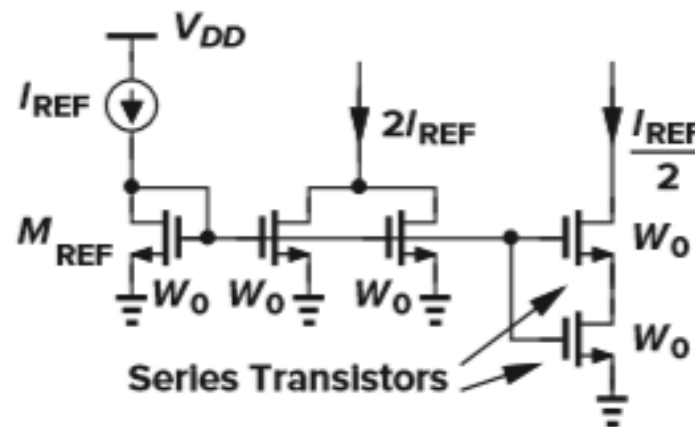
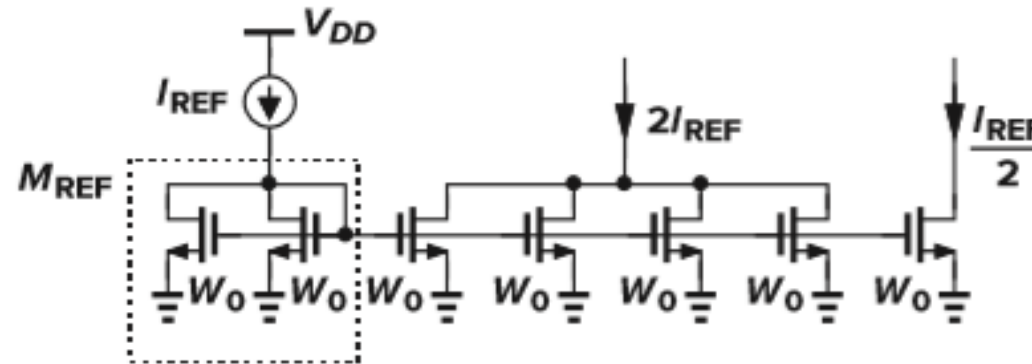
Practical Implementations

- Gate corner leads to inaccurate sizes
 - Multi-fingers with same sizes



$2*I_{REF}$ & $I_{REF}/2$

- Two methods for realizing $2.5I_{REF}$



Outline

- Basic current mirrors
- **Cascode current mirrors**
- Active current mirrors
- Bias techniques



Drawback of the basic current mirror

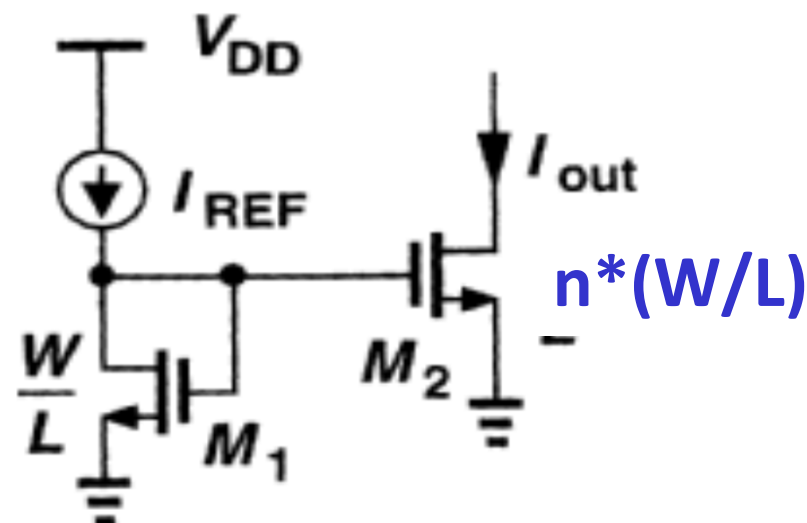
- $\lambda > 0$, resulting in significant error in copying I_{REF}

$$I_{REF} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_1 (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS1})$$

$$I_{out} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_2 (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS2})$$

$$\frac{I_{out}}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1} \cdot \frac{1 + \lambda V_{DS2}}{1 + \lambda V_{DS1}}$$

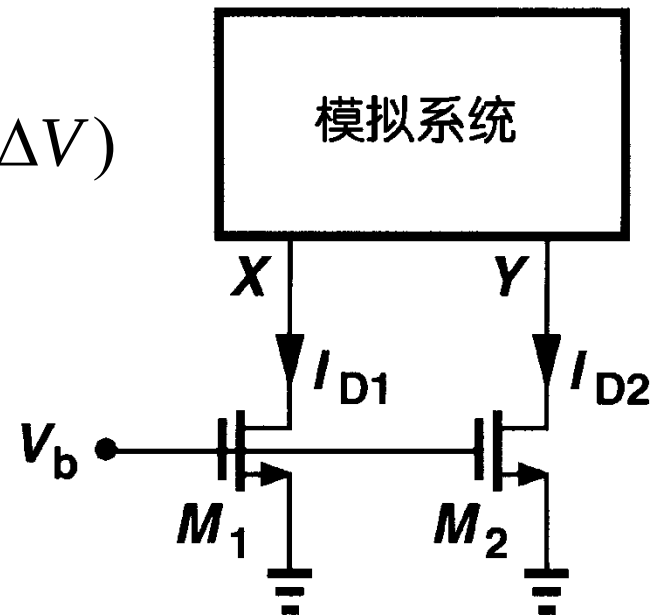
$$V_{DS1} = V_{GS1} \neq V_{DS2}$$



Understanding the shielding property of cascode

- Considering current source array case
 - M1 and M2 are designed with the same size
 - Ideally, $I_{D1}=I_{D2}$
 - $V_X-V_Y=\Delta V$ because of the analog system

$$I_{D1} - I_{D2} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_b - V_{TH})^2 (\lambda \cdot \Delta V)$$



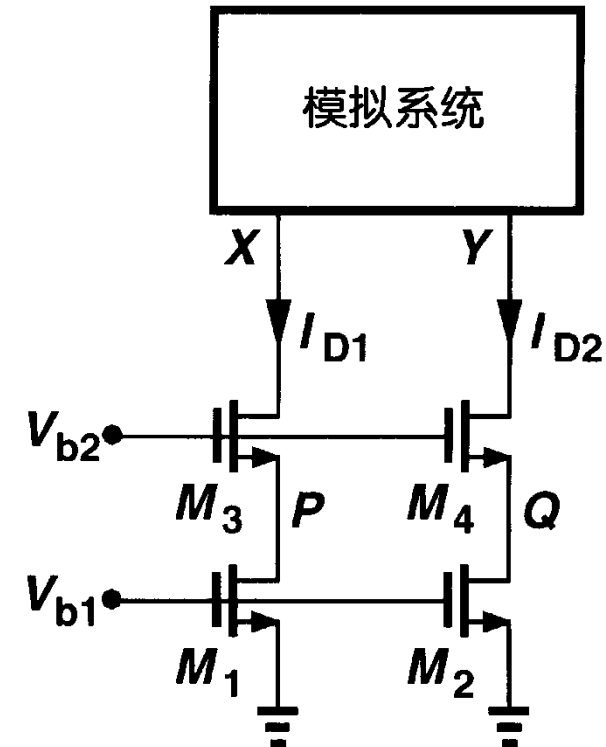
Cascode case

- $V_X - V_Y = \Delta V$

$$\Delta I = \frac{\Delta V}{(g_{m3,4} + g_{mb3,4})r_{o3,4} \cdot r_{o1,2}}$$

$$\Delta V_{P,Q} = \Delta I \cdot r_{o1,2} = \frac{\Delta V}{(g_{m3,4} + g_{mb3,4})r_{o3,4}}$$

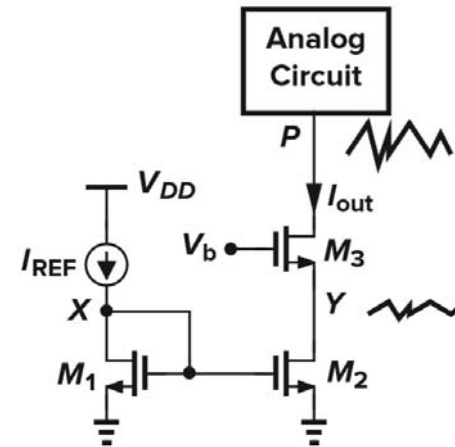
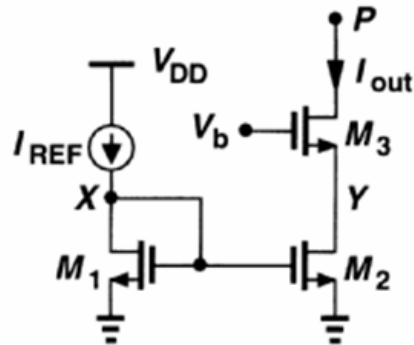
$$I_{D1} - I_{D2} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_b - V_{TH})^2 \cdot \frac{\lambda \cdot \Delta V}{(g_{m3,4} + g_{mb3,4})r_{o3,4}}$$



- The variations at node X/Y are shielded by M3,4 !**

Improvement: cascode current source

- Use cascode to reject channel length modulation



$$\Delta V_Y \approx \Delta V_P / [(g_{m3} + g_{mb3})r_{O3}]$$

- Chose V_b to ensure $V_Y = V_X \rightarrow I_{out}$ tracks I_{REF} closely

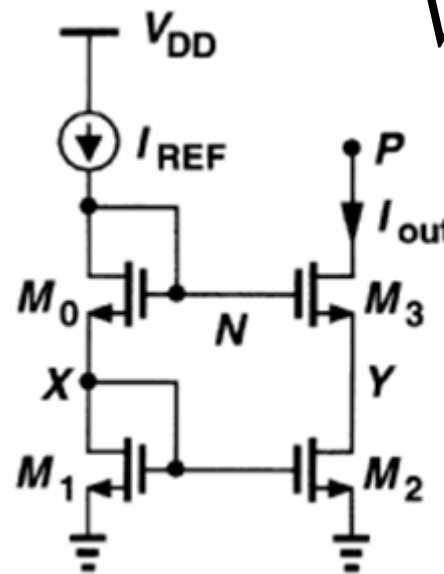
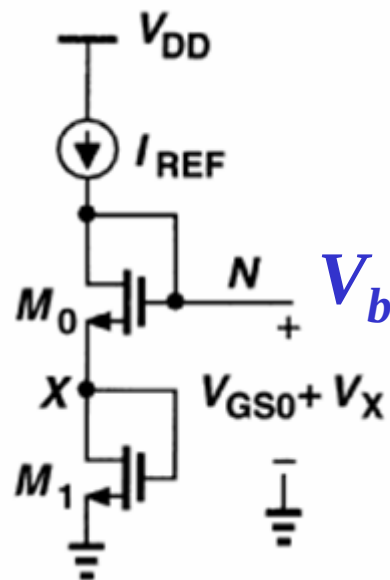
we require: $V_b - V_{GS3} = V_X$

- How to generate V_b ?



Cascode current mirror

- Modify the mirror to generate V_b
 - Add another diode-connected NMOS (M0)



$$V_{GS0} + V_X = V_{GS3} + V_Y$$

if

$$\frac{(W/L)_2}{(W/L)_1} = \frac{(W/L)_3}{(W/L)_0}$$

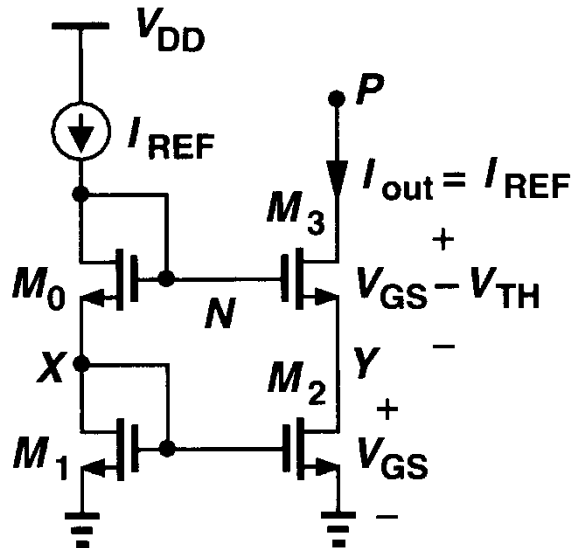
..... $V_{GS0} = V_{GS3}$

..... $V_X = V_Y$

Minimum output voltage of cascode mirror

- To ensure all MOS operate in saturation region
- The minimum voltage of V_P is

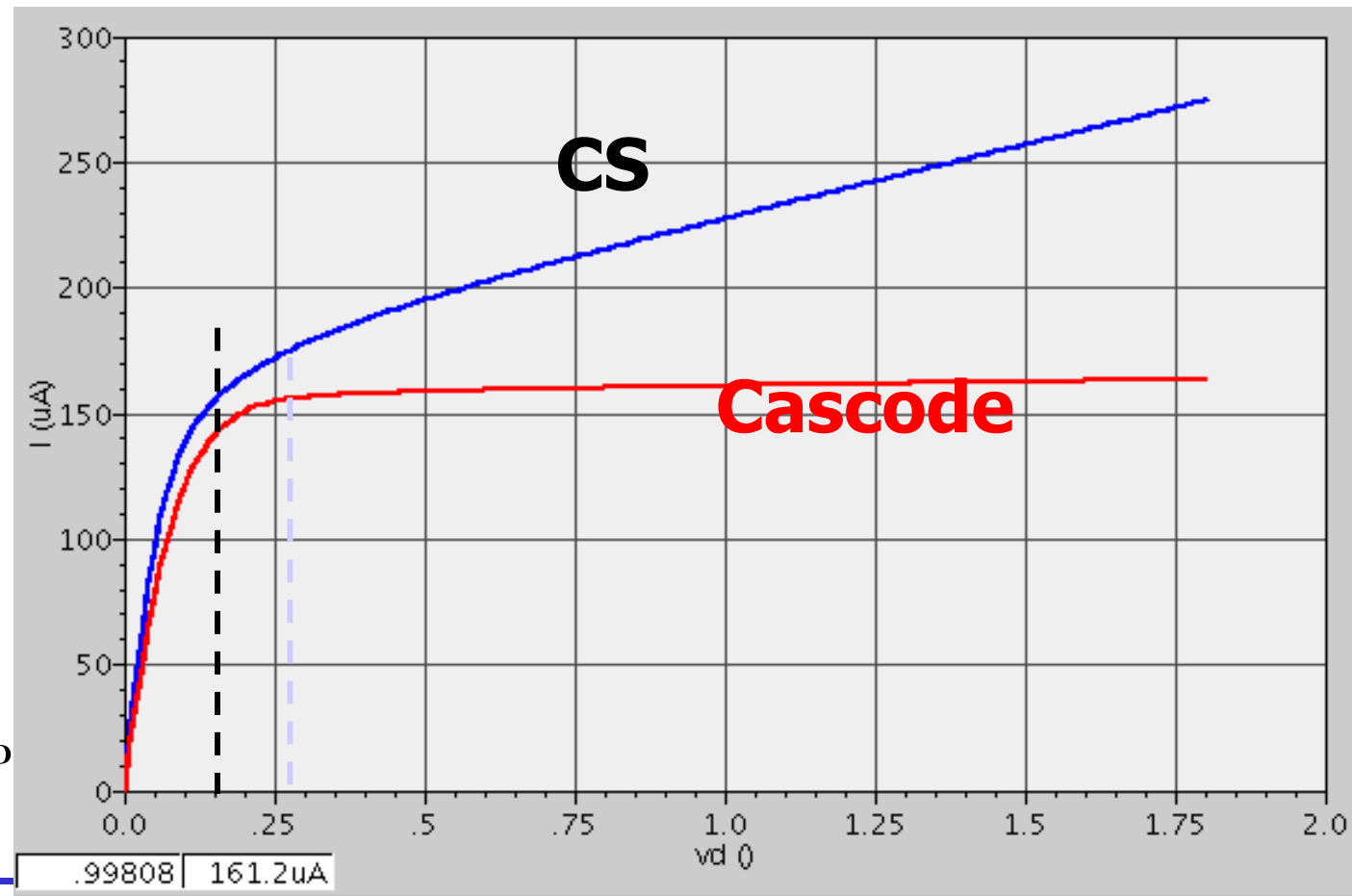
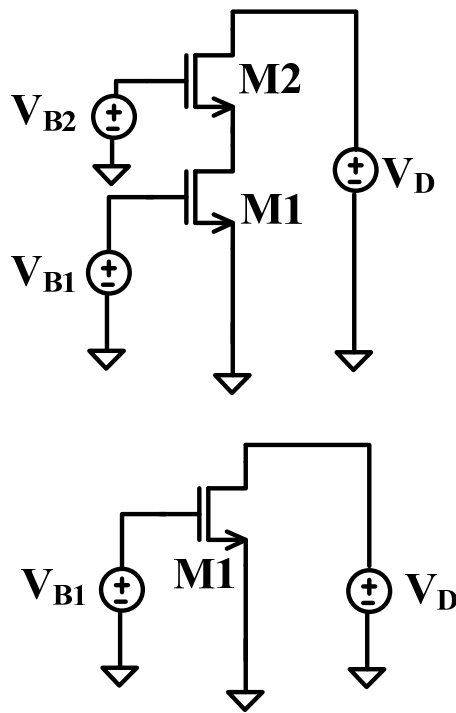
$$\begin{aligned} V_{P,\min} &= V_N - V_{TH} = V_{GS1} + V_{GS0} - V_{TH} \\ &= (V_{GS0} - V_{TH}) + (V_{GS1} - V_{TH}) + V_{TH} \\ &= 2V_{OD} + V_{TH} \end{aligned}$$



Comparison of CS and Cascode

Sizes: M1, M2: $W=10\mu$, $L=0.18\mu$

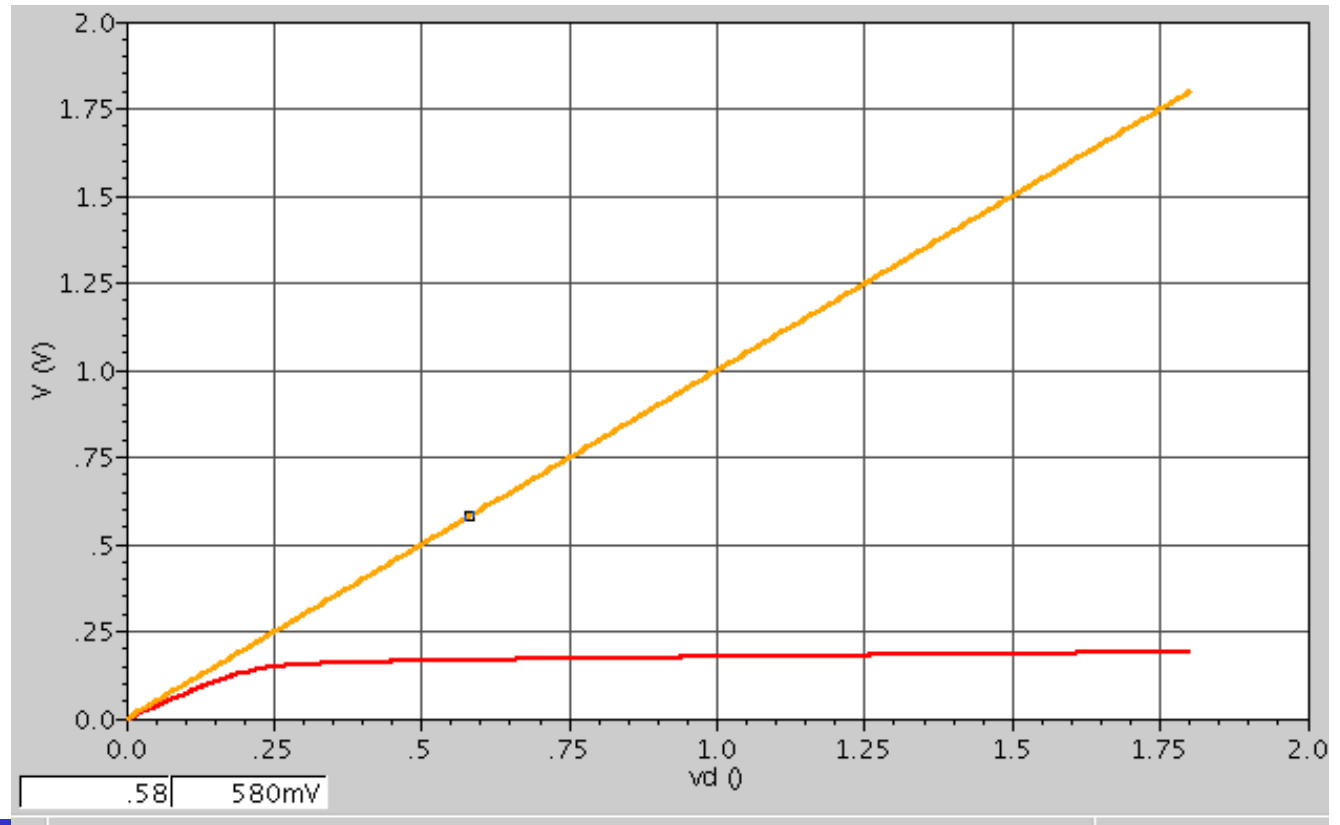
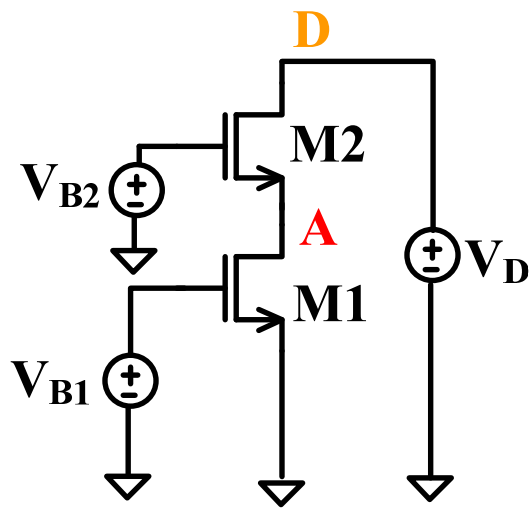
Bias Voltage: $V_{B1}=0.6$, $V_{B2}=0.8$; $V_D=0\sim 1.8$



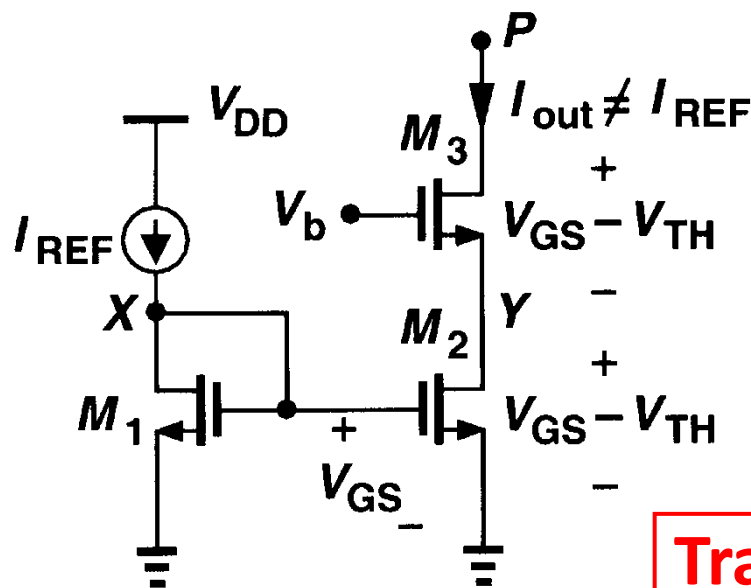
Cascode Node votages

Variation of the voltage at node D is shielded by M2
Leading to small variation at node A

→ Stabilizing current of M1



- If we do not require $V_X = V_Y$ ($I_{out} \neq I_{REF}$)
 - V_b can be set to a lower value, permitting a lower minimum V_P



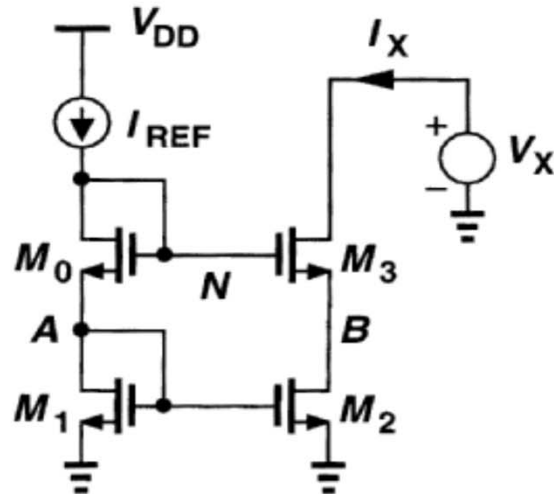
$$\begin{aligned}
 V_b &= V_{GS1} - V_{TH} + V_{GS3} \\
 \dots \rightarrow V_{P,min} &= V_b - V_{TH} \\
 &= V_{GS1} - V_{TH} + V_{GS3} - V_{TH} \\
 &= 2V_{OD}
 \end{aligned}$$

Trade off: Accuracy—Headroom

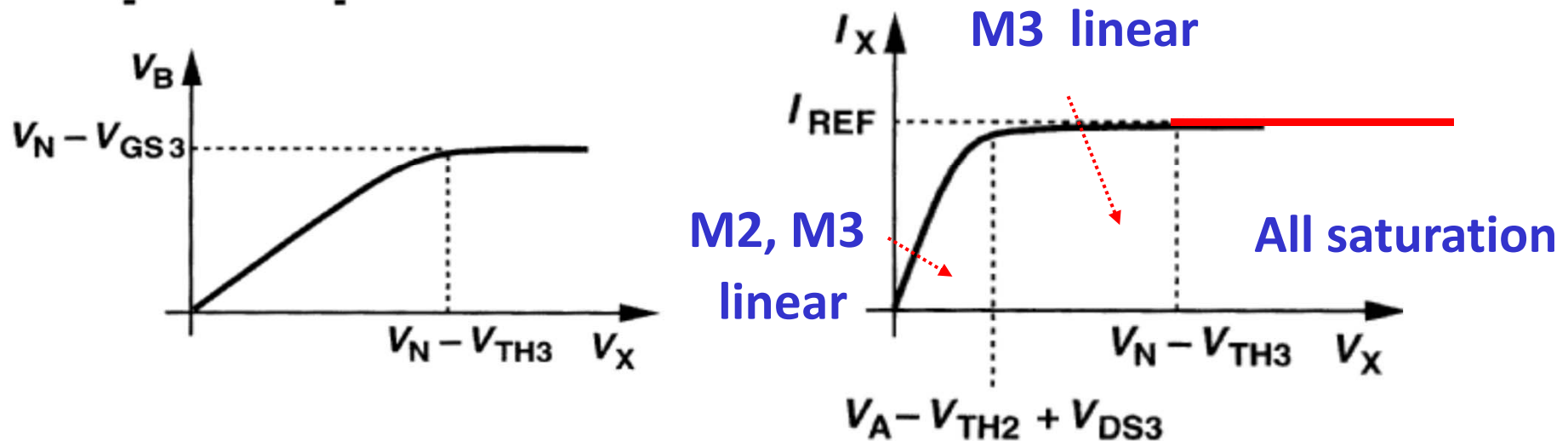
- The cascode mirror achieves high accuracy at the price of wasting a V_{TH} in the headroom !

Ex. 5.4 Cascode Mirrors curves

- Sketch I_X and V_B curves as V_X drops from a large value



- When V_X is larger than $V_N - V_{TH3}$
 - $I_X = I_{REF}$, $V_B = V_N - V_{GS3}$
- When V_X is lower than $V_N - V_{TH3}$
 - M3 enters linear region first
 - V_B drops, I_X drops.



Low-voltage cascode mirror (1)

- A cascode topology with output shorted to input

– How to set V_b ?

$$V_X = V_{GS1}$$

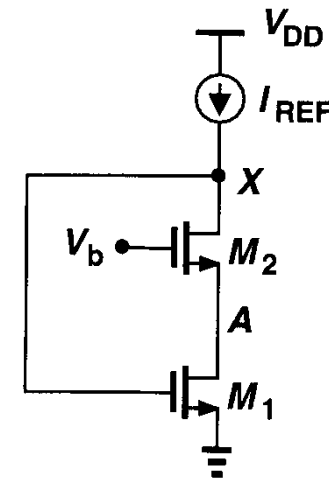
- For M2 to be saturated:

$$V_b - V_{TH2} \leq V_X = V_{GS1}$$

- For M1 to be saturated:

$$V_{GS1} - V_{TH1} \leq V_A = V_b - V_{GS2}$$

→ $V_{GS2} + (V_{GS1} - V_{TH1}) \leq V_b \leq V_{GS1} + V_{TH2}$



– If $V_{GS1} - V_{TH1} < V_{TH2}$, above expression holds



Low-voltage cascode mirror (2)

- If
$$\frac{(W/L)_2}{(W/L)_4} = \frac{(W/L)_1}{(W/L)_3}$$

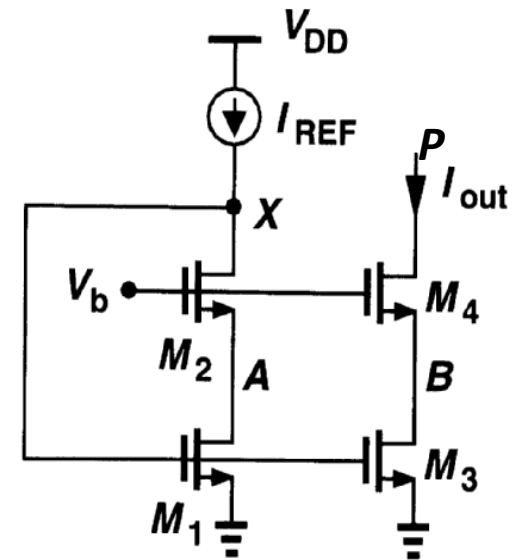
..... $V_{GS2} = V_{GS4}$ $V_A = V_B$

– Copying accuracy is achieved

- When V_b is at its minimum

$$V_b = V_{GS2} + (V_{GS1} - V_{TH1})$$

$$\begin{aligned} V_{P,\min} &= V_b - V_{TH} = (V_{GS2} - V_{TH}) + (V_{GS1} - V_{TH1}) \\ &= 2V_{OD} \end{aligned}$$



Circuits to generate V_b (1)

- For minimum headroom consumption, we need:

$$V_A = V_{GS1} - V_{TH1} \rightarrow V_b = V_{GS2} + V_{GS1} - V_{TH1}$$

- M_5 , M_6 together with R_b generates V_b

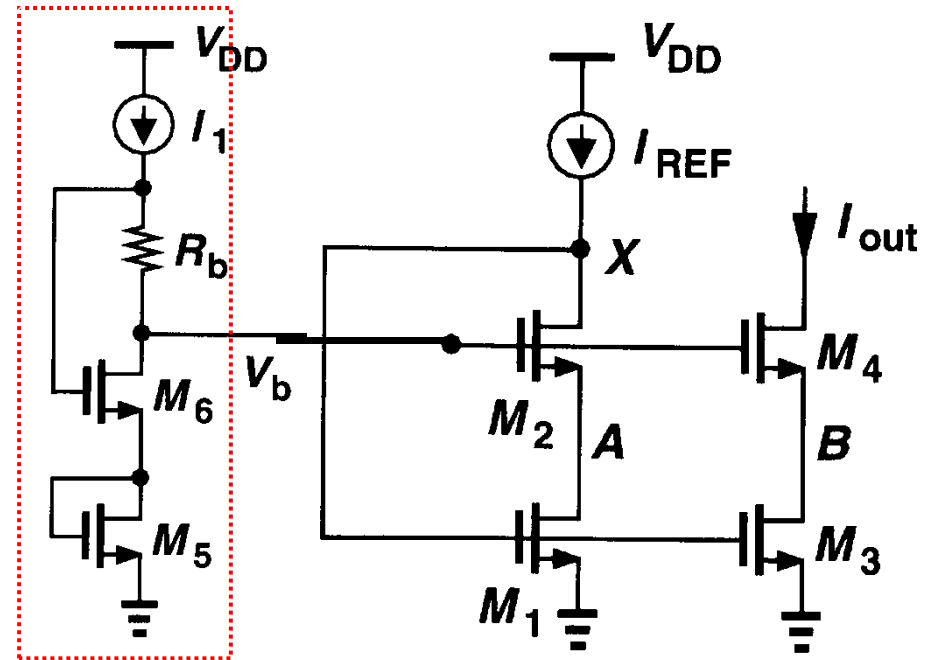
$$- V_{GS5} \approx V_{GS2}, V_{DS6} \approx V_{GS6} - I_1 R_b$$

- Proper design makes

$$- I_1 R_b \approx V_{TH1}$$

$$\rightarrow V_b \approx V_{GS2} + V_{GS1} - V_{TH1}$$

- Not accurate**
 - Resistor
 - Body effect



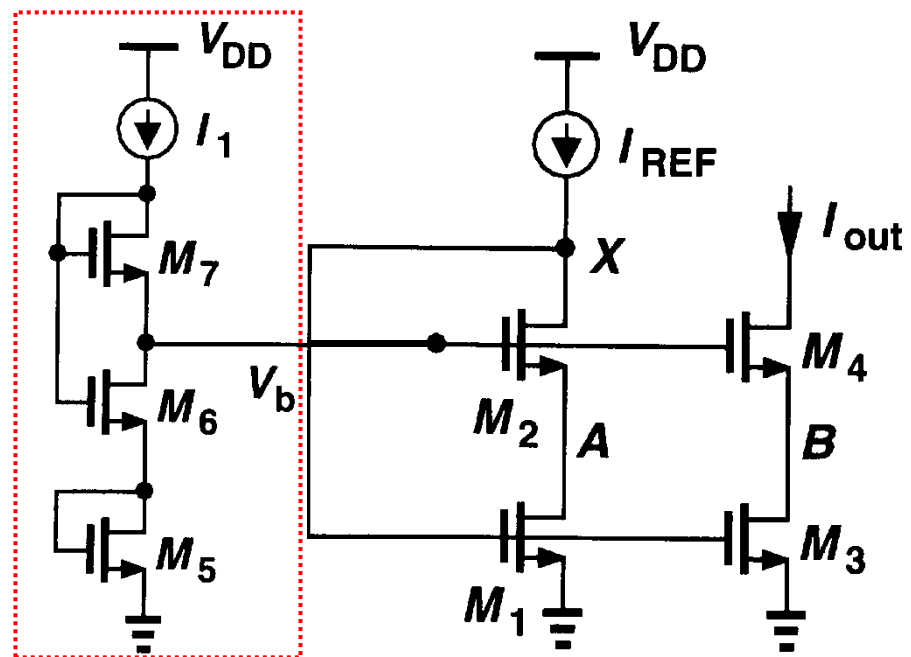
Circuits to generate V_b (2)

- M7 with a large enough W/L , so that

$$V_{GS7} \approx V_{TH7}$$

$$\longrightarrow V_{DS6} \approx V_{GS6} - V_{TH7} \longrightarrow V_b \approx V_{GS5} + V_{GS6} - V_{TH7}$$

- Without Resistor
- Still suffer from body effect



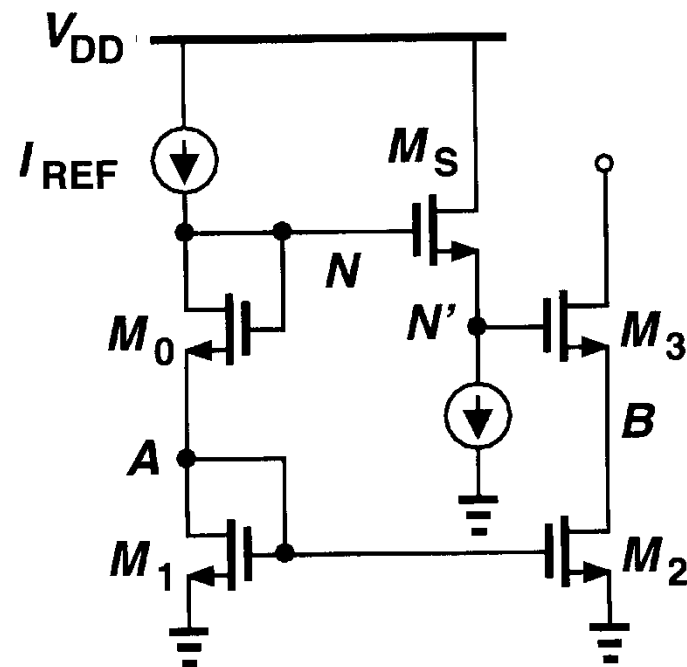
Circuits to generate V_B (3)

- Use a source follower to generate biasing voltage
 - M_S is biased with very small current

$$V_{GS,MS} \approx V_{TH,MS} \quad V_{N'} \approx V_N - V_{TH,MS} \approx V_N - V_{TH3}$$

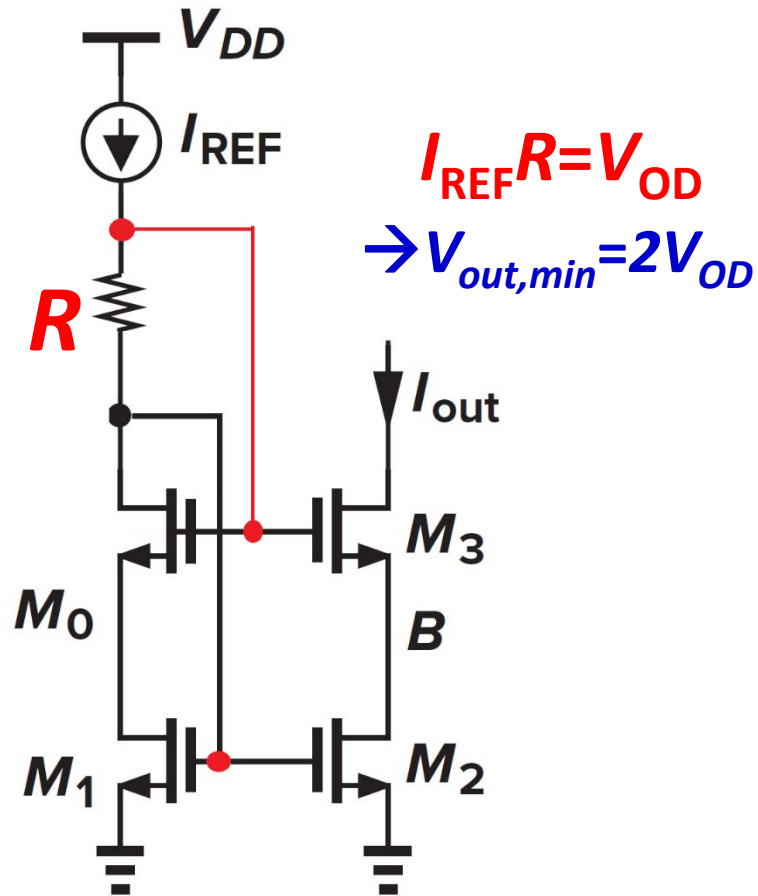
$$V_B \approx V_{GS1} + V_{GS0} - V_{TH3} - V_{GS3} \\ \approx V_{GS1} - V_{TH3}$$

- $V_A \neq V_B$
 - Not accurate

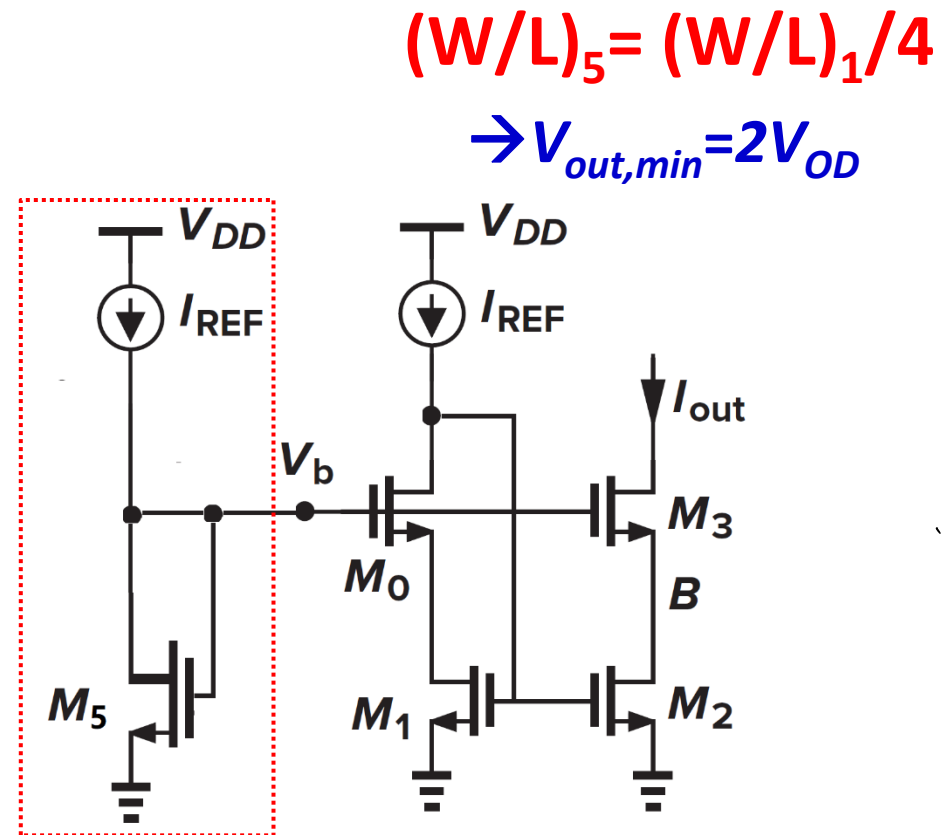


More feasible solutions

- Self-biased Cascode

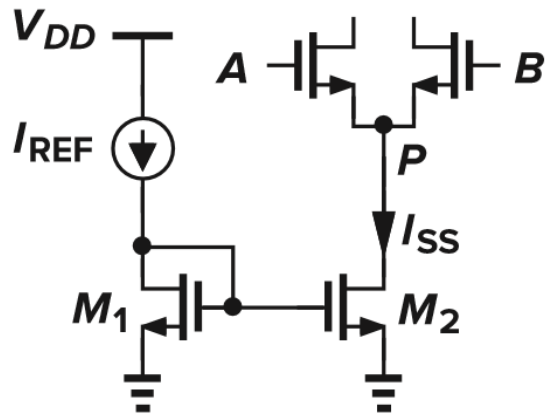


- Stacked MOS
(Fig. 5.19(b))



Example 5.6

- How to reduce the current error resulted from channel length modulation without using cascode?

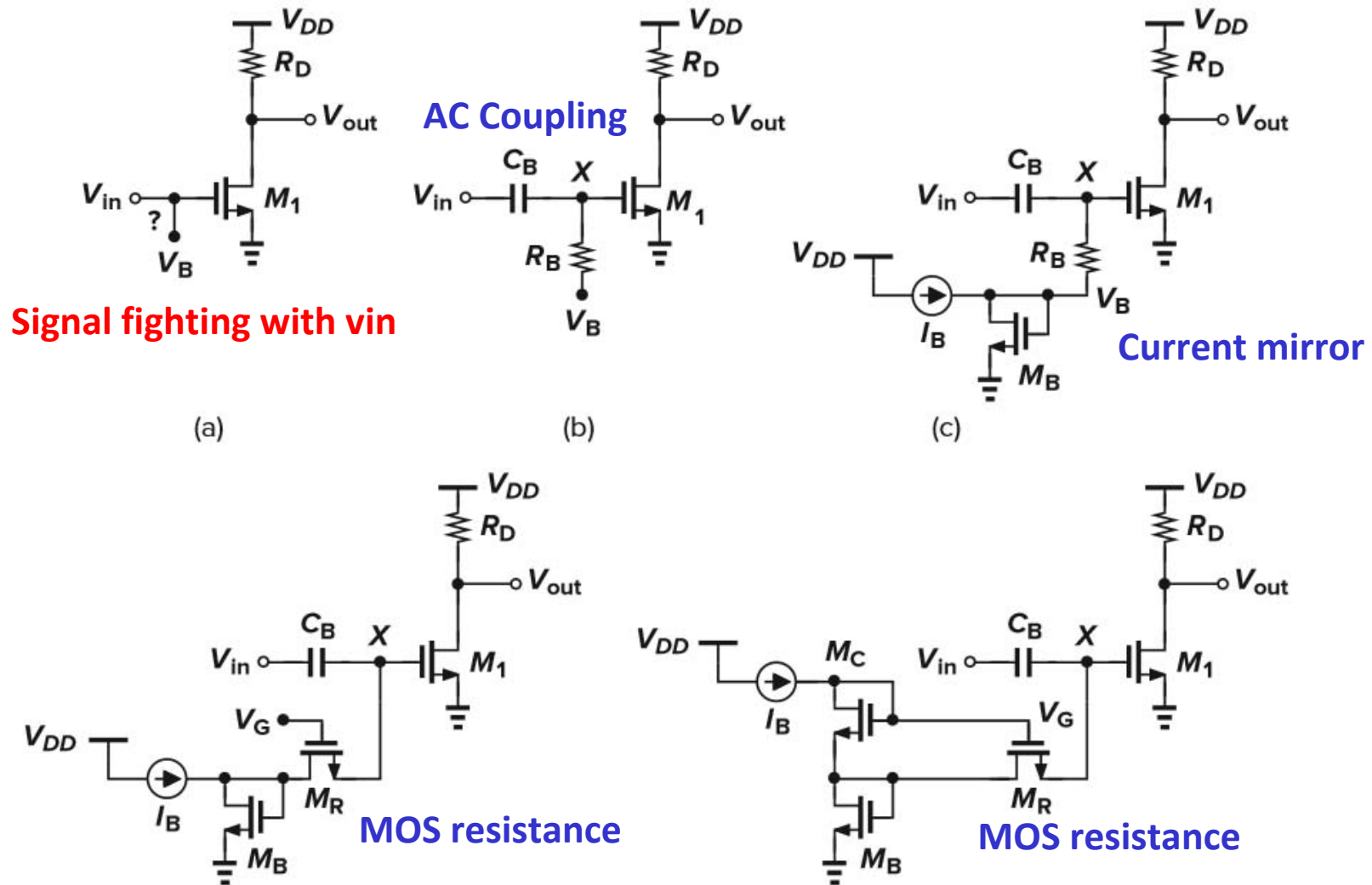


Outline

- Basic current mirrors
- Cascode current mirrors
- Active current mirrors
- **Bias techniques**



Bias for CS stage



CS-stage with current source load

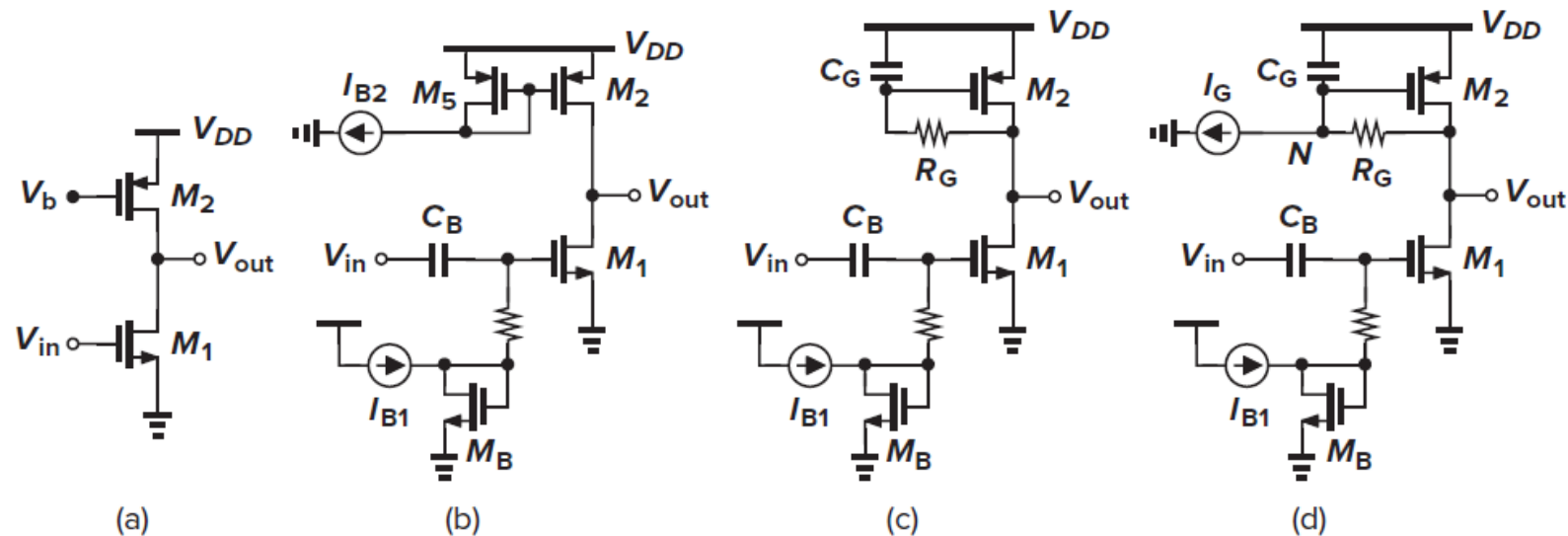


Figure 5.45 (a) CS stage with current-source load; (b) simple biasing; (c) self-biasing of current source; (d) use of I_G to shift the output.

- **Self-biased gain:** $A_v = -g_{m1}(r_{O1} || r_{O2} || R_G)$

R_G needs to be large to ensure high gain

Bias for CG stage

$$R_B/R_S = I_{D1}/I_B$$

Current Mirror

Low-voltage
Current Mirror

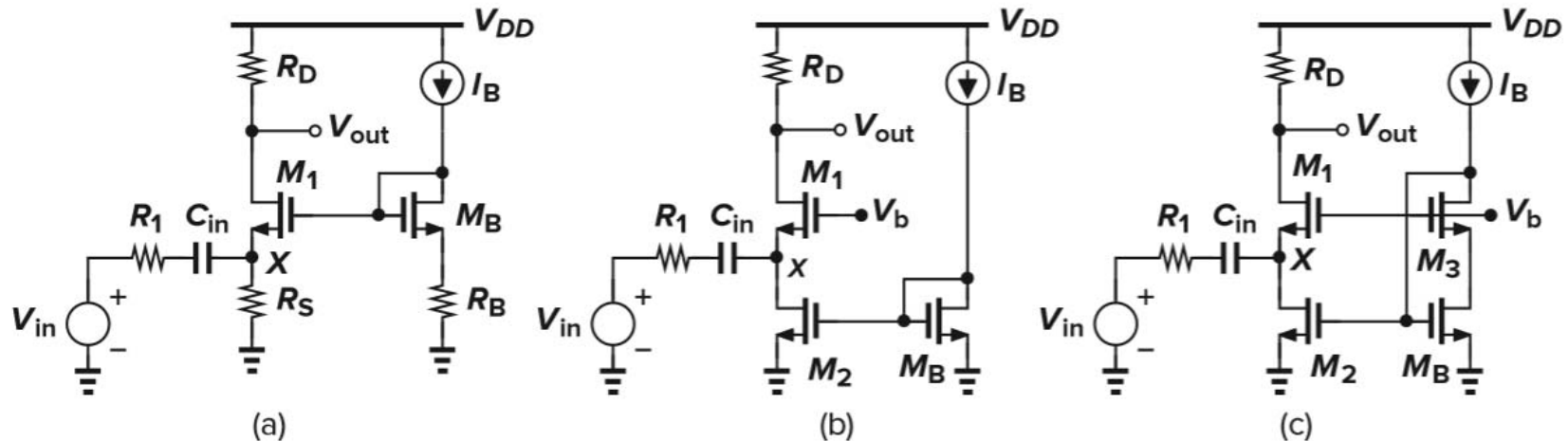
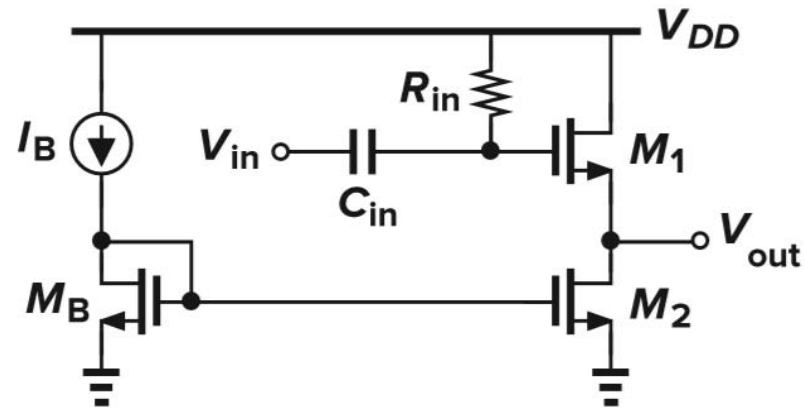
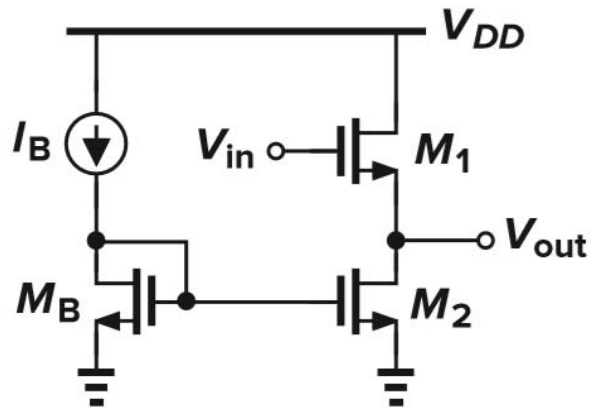


Figure 5.48 CG stage with (a) resistive path from source to ground, (b) current-source biasing, and (c) low-voltage current mirror.

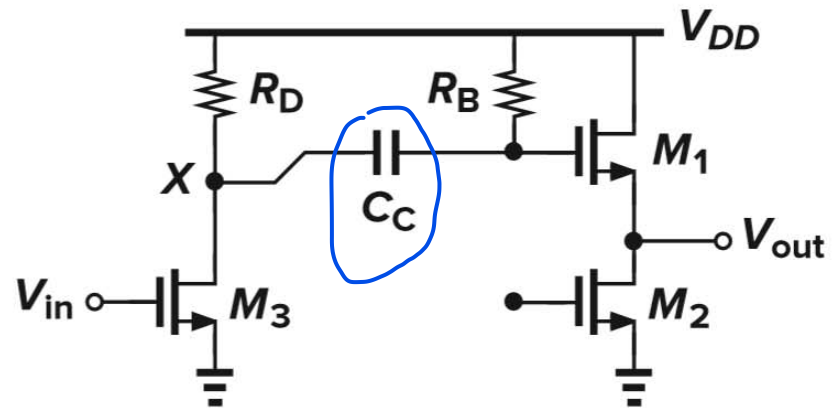
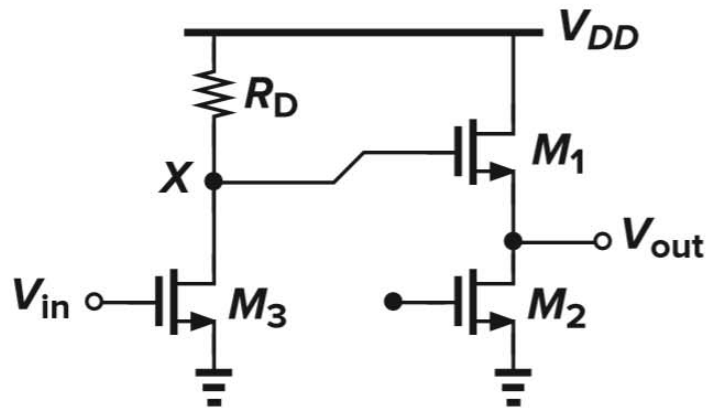


Bias for SF

- SF bias



- CS + SF



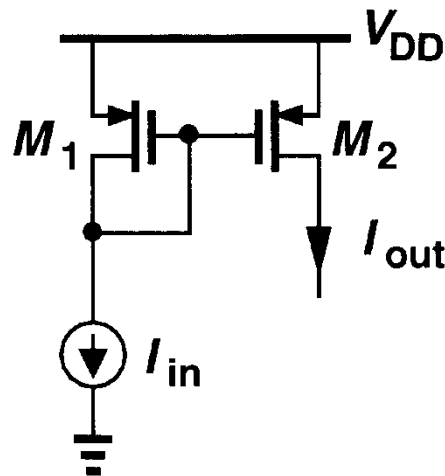
Outline

- Basic current mirrors
- Cascode current mirrors
- **Active current mirrors**
- Bias techniques



Current mirror as signal processing block

- **M1 and M2 are identical**
 - If $\lambda=0$, $I_{out} = I_{in} \rightarrow i_{out}=i_{in}$
 - Performs signal processing without inversion



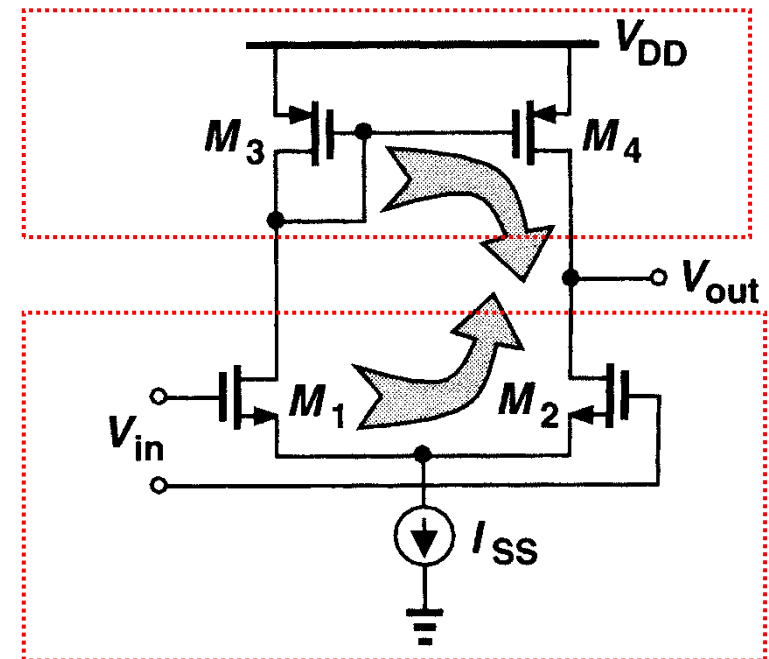
- It can be used as the load of differential pairs



Use current mirror as the loads of diff. pair

- Also called active current mirror
- Convert differential output to single-ended
- When $V_{GS1} \uparrow$ ($V_{GS2} \downarrow$),
 - $\rightarrow I_{D1} \uparrow$ ($I_{D2} \downarrow$)
 - Because of Mirror
 - $I_{D4} \uparrow$
 - V_{out} increases sharply
 - Gain is enhanced by the mirror
- Very useful analog block

Active current mirror

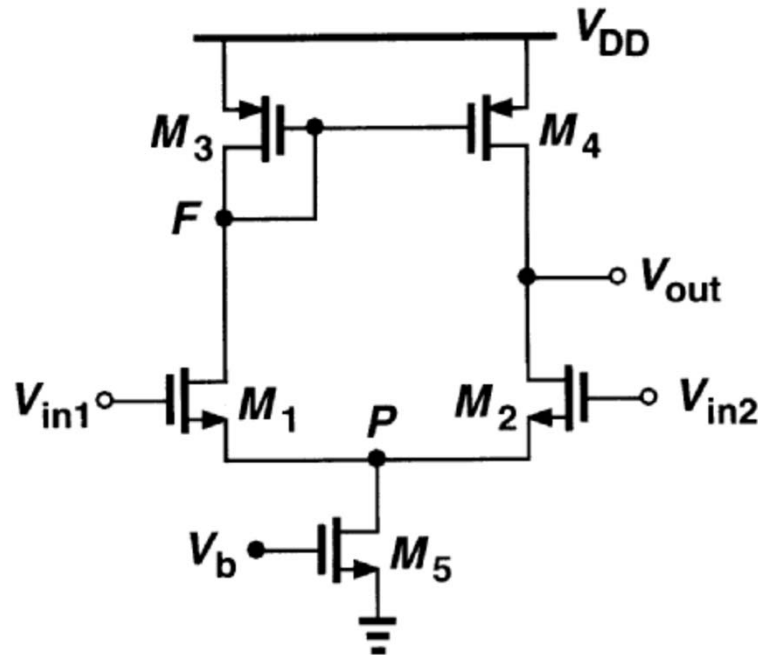


Diff. pair



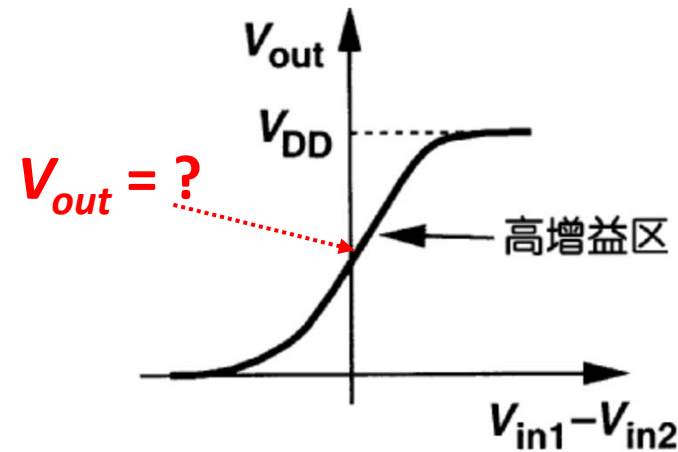
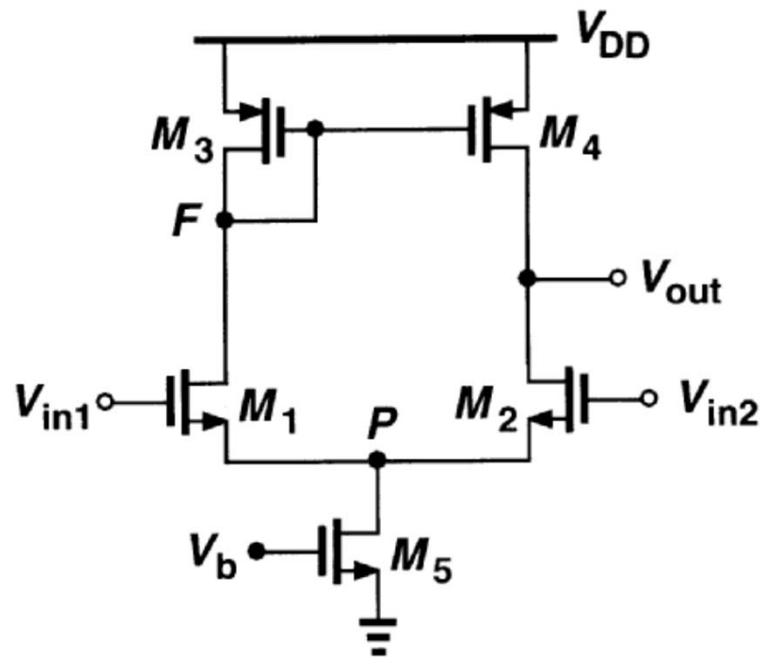
Large-signal behavior

- **Considering $V_{in1}-V_{in2}$ increases from negative to positive value**
 - **What is the output voltage when $V_{in1}=V_{in2}$?**
- **Determine the input common-mode range**



Large-signal behavior: **diff. in $\rightarrow$ single-ended output**

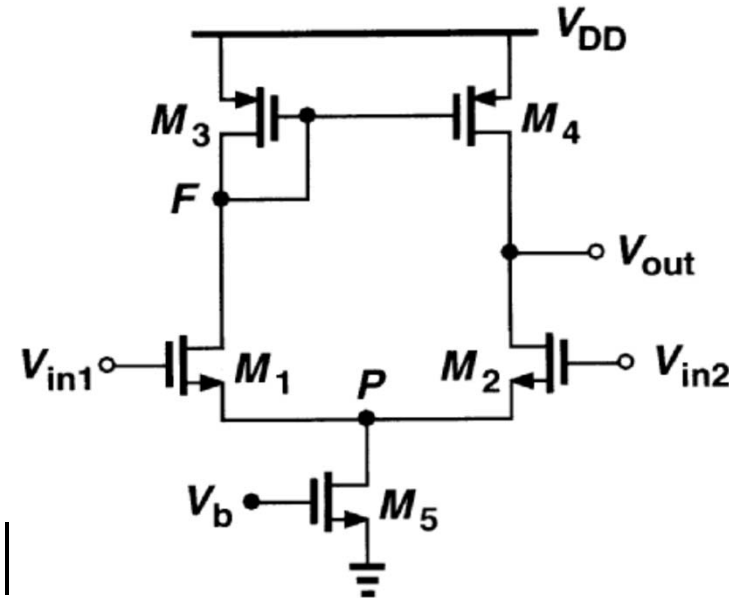
- Considering $V_{in1} - V_{in2}$ increases from negative to positive value



What is the output voltage when $V_{in1}=V_{in2}$?

- When $V_{in1}=V_{in2}$
 - $V_F = V_{DD} - |V_{GS3}|$
 - Because of symmetry

$$\begin{aligned} V_{out} \big|_{\text{when } V_{in1}=V_{in2}} &= V_F \\ &= V_{DD} - |V_{GS3}| \end{aligned}$$



when $V_{in1}=V_{in2}$,

$V_{out}=V_F$ is the unique solution

The output DC point is well defined



Large-signal behavior: **Input common-mode range**

– Input common-mode range

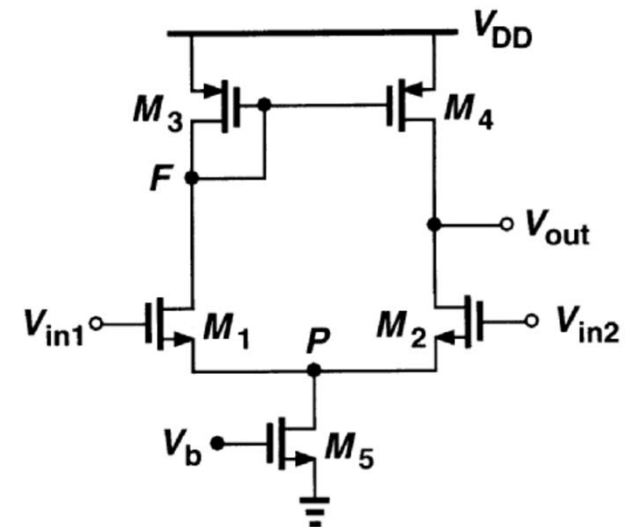
$$V_{OD5} + V_{GS1} \leq V_{in,CM} \leq V_{out} + V_{TH}$$

$$V_{OD5} + V_{GS1} \leq V_{in,CM} \leq V_F + V_{TH}$$

$$V_F = V_{DD} - |V_{GS3}|$$

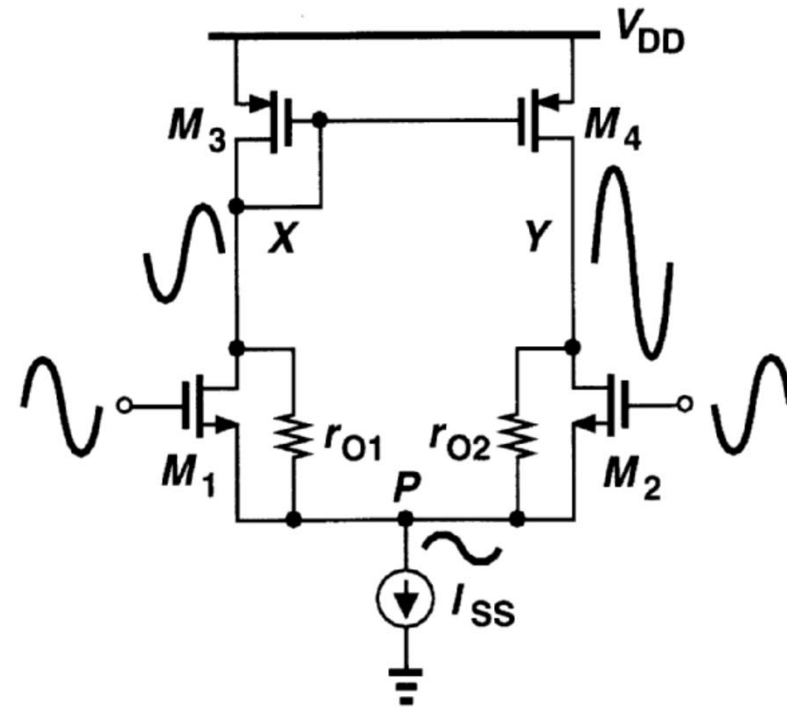
Output range is influenced by $V_{in,CM}$

--- A drawback



Small-Signal Analysis

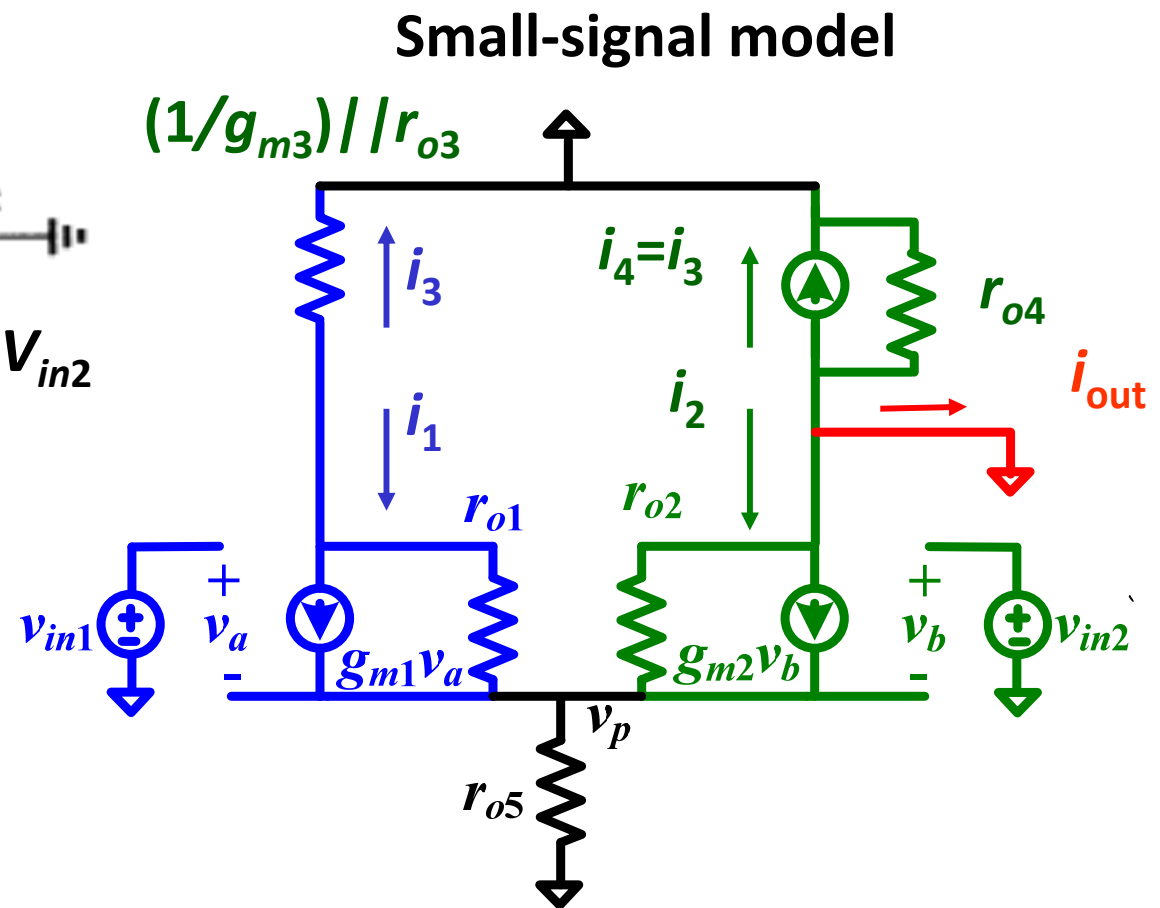
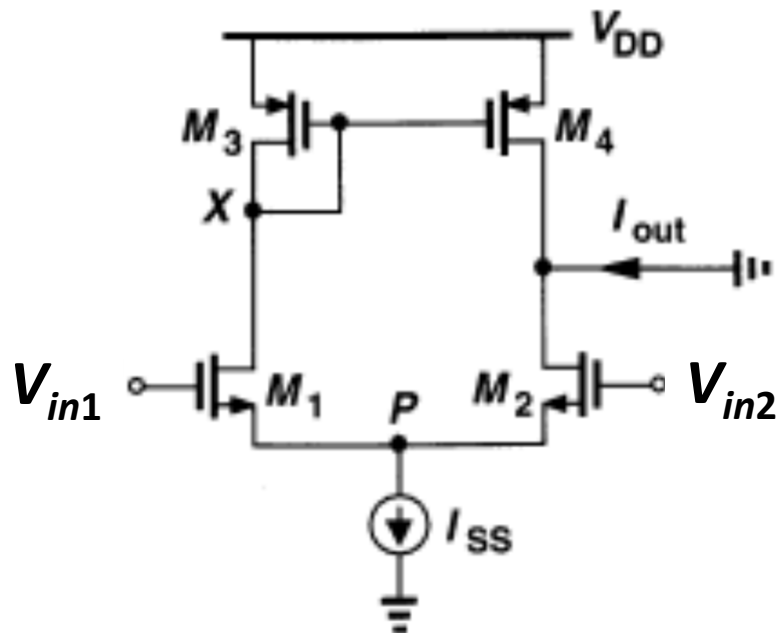
- Can we use half-circuit?
- Because impedance at X is much smaller than Y
 - Effects of V_X and V_Y at P do not cancel
 - Node P is not virtual ground
 - Half-circuit concept can not be used
- Calculate G_m and r_{out} separately



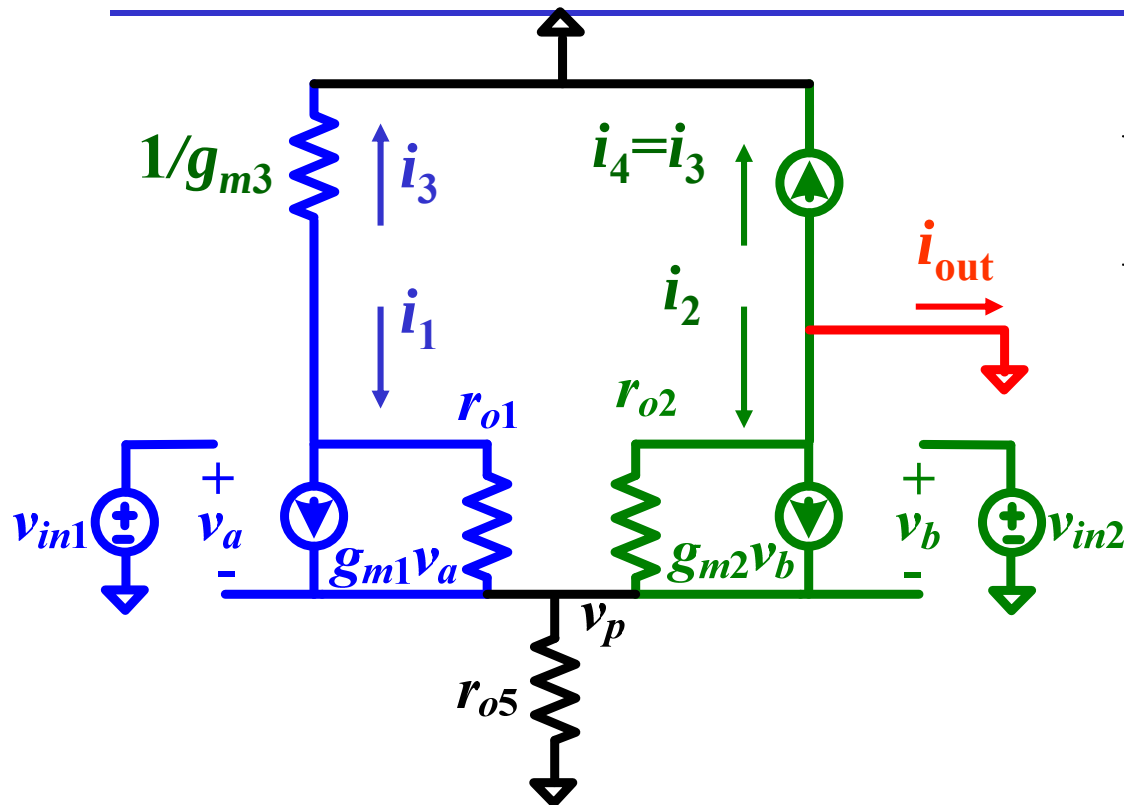
$$A_v = G_m \cdot r_{out}$$

Calculation of G_m

Output shorted, calculating small-signal current



Calculation of G_m



$$V_{id} = V_{in1} - V_{in2}$$

$$V_{ic} = (V_{in1} + V_{in2}) / 2 = V_p$$

$$i_3 = -i_1 \quad i_4 = i_3$$

$$i_{out} = -i_4 - i_2$$

$$i_1 = g_{m1}(V_{in1} - V_p) = g_{m1} \frac{V_{id}}{2} \quad (g_{m1} = g_{m2}) \quad i_{out} = g_{m1,2} V_{id}$$

$$i_2 = g_{m2}(V_{in2} - V_p) = -g_{m2} \frac{V_{id}}{2}$$

$$G_m = g_{m1,2}$$



Calculation of r_{out}

Apply a test voltage v_t , calculate $i_t \rightarrow r_{out} = v_t / i_t$

Decomposing it into four parts $i_{t1} \sim i_{t4}$

$$i_{t4} = v_t / r_{o4}$$

$$r_{eq} \approx g_{m2} r_{o2} \cdot (1 / g_{m1}) + r_{o2} = 2r_{o2}$$

$$i_{t1} + i_{t2} = v_t / r_{eq} = v_t / 2r_{o2}$$

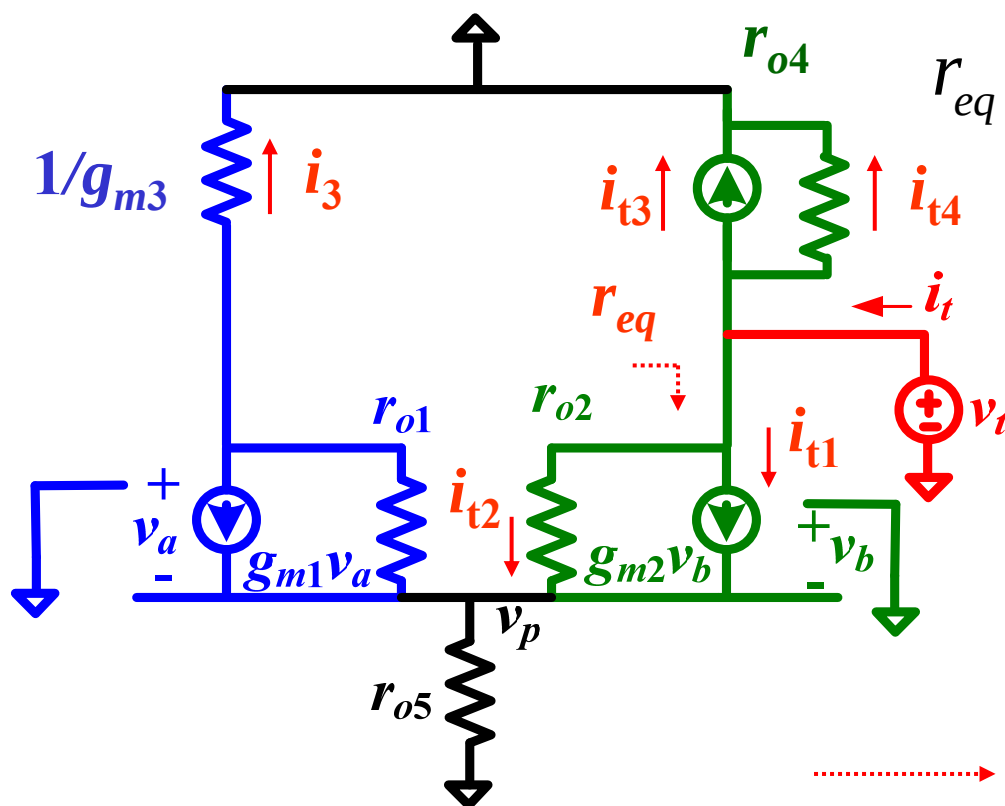
$$i_{t3} = i_3 = i_{t1} + i_{t2} = v_t / 2r_{o2}$$

mirror

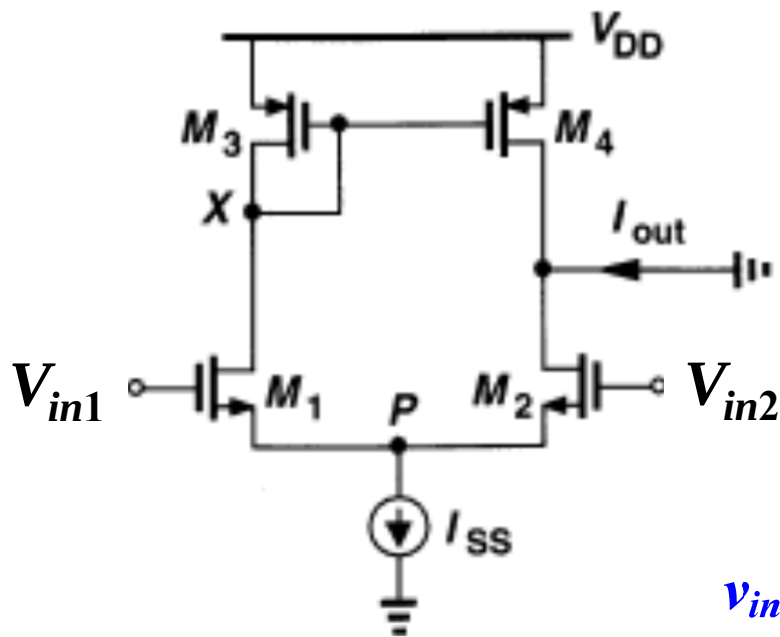
$$i_t = i_{t1} + i_{t2} + i_{t3} + i_{t4}$$

$$= v_t / 2r_{o2} + v_t / 2r_{o2} + v_t / r_{o4}$$

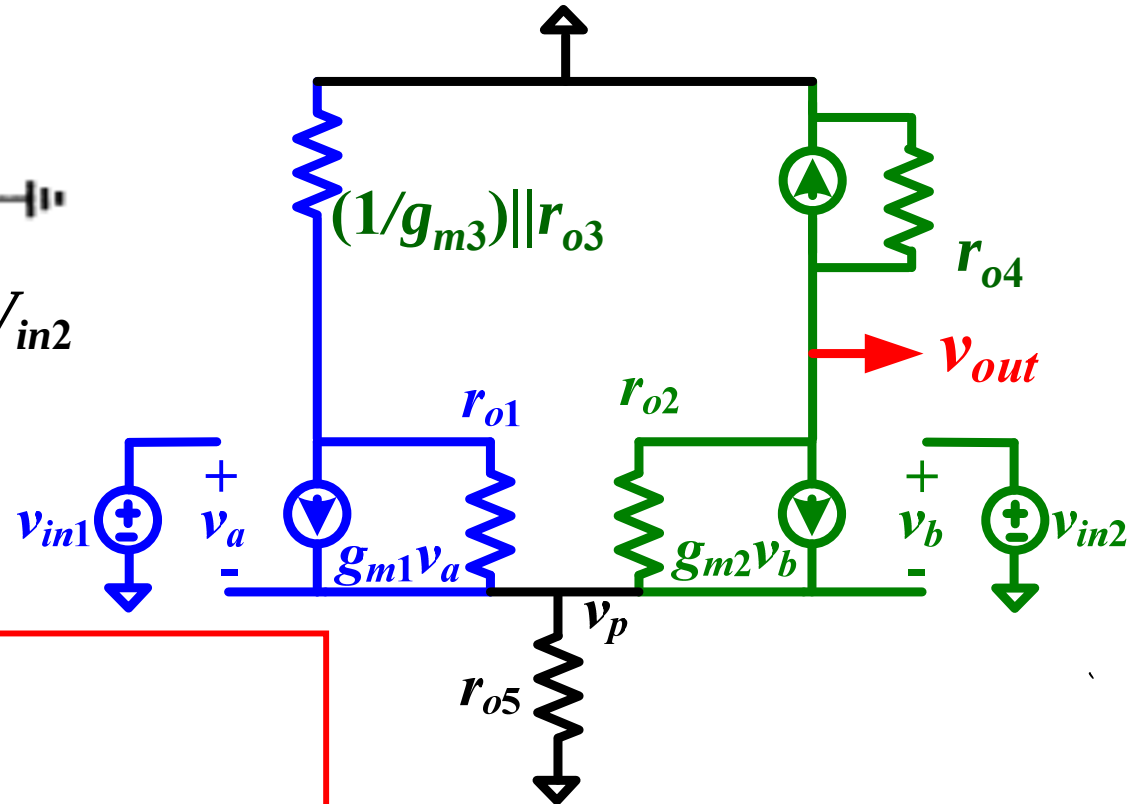
$$r_{out} = v_t / i_t \approx r_{o2} \parallel r_{o4}$$



Small-signal gain



Small-signal model



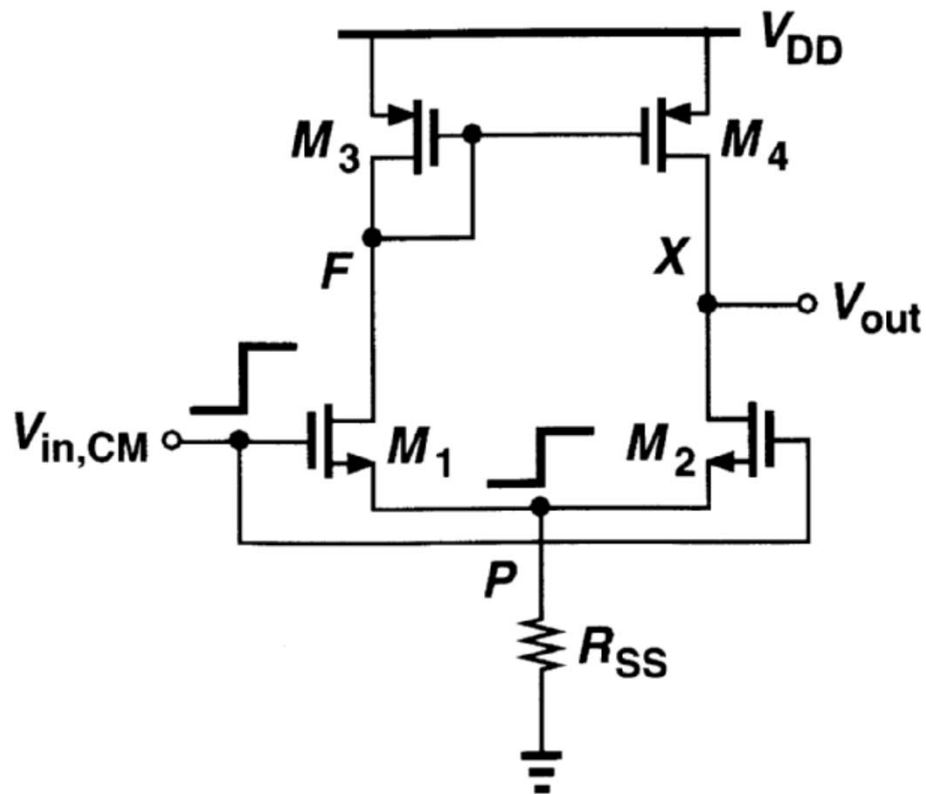
$$A_V = \frac{v_{out}}{v_{id}} = \frac{v_{out}}{v_{in1} - v_{in2}}$$

$$= G_m \cdot r_{out} = g_{m1,2} (r_{o2} \parallel r_{o4})$$



Common Mode Characteristics

Finding the effects of variation in $V_{in,CM}$ on V_{out}
considering only the situation of finite tail resistance

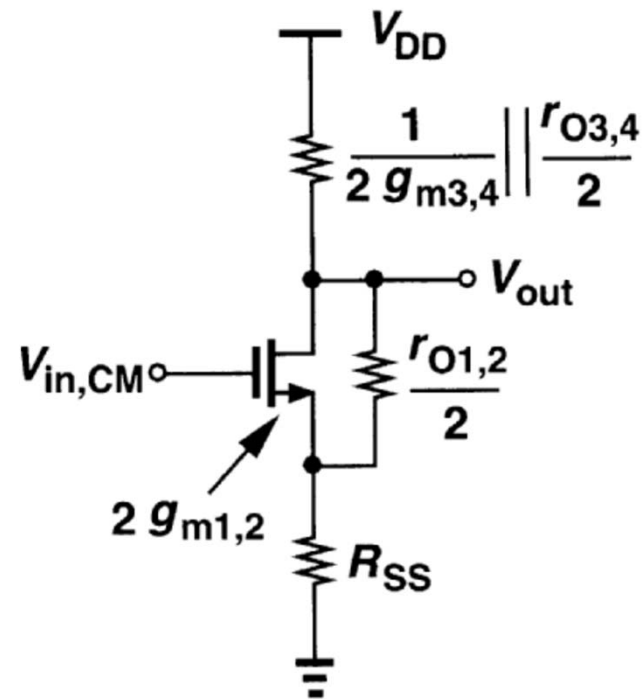
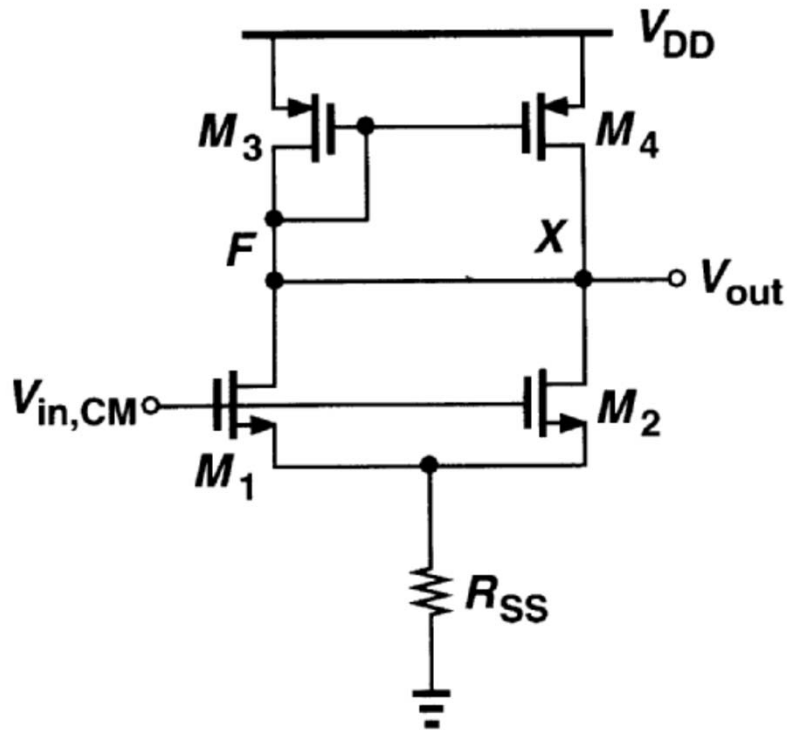


$$A_{CM} = \frac{\Delta V_{out}}{\Delta V_{in,CM}}$$

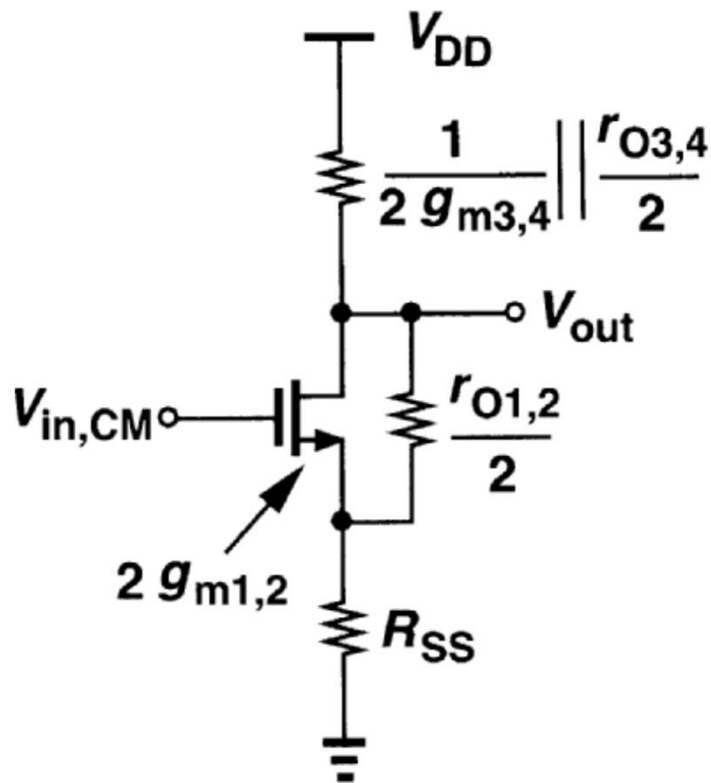


Calculation of A_{CM}

- Because of symmetry, V_{out} always equals to V_F
 - Connect the two nodes together



- Essentially a CS stage with source degeneration



$$A_{CM} = \frac{\Delta V_{out}}{\Delta V_{in,CM}} \approx \frac{-\frac{1}{2g_{m3,4}} \parallel \frac{r_{o3,4}}{2}}{\frac{1}{2g_{m1,2}} + R_{SS}}$$

$$\approx \frac{-1}{1 + 2g_{m1,2}R_{SS}} \frac{g_{m1,2}}{g_{m3,4}}$$



CMRR

- Common-mode rejection ratio

$$CMRR = \left| \frac{A_{DM}}{A_{CM}} \right| = g_{m1,2} (r_{o1,2} \parallel r_{o3,4}) \frac{g_{m3,4} (1 + 2g_{m1,2} R_{SS})}{g_{m1,2}}$$

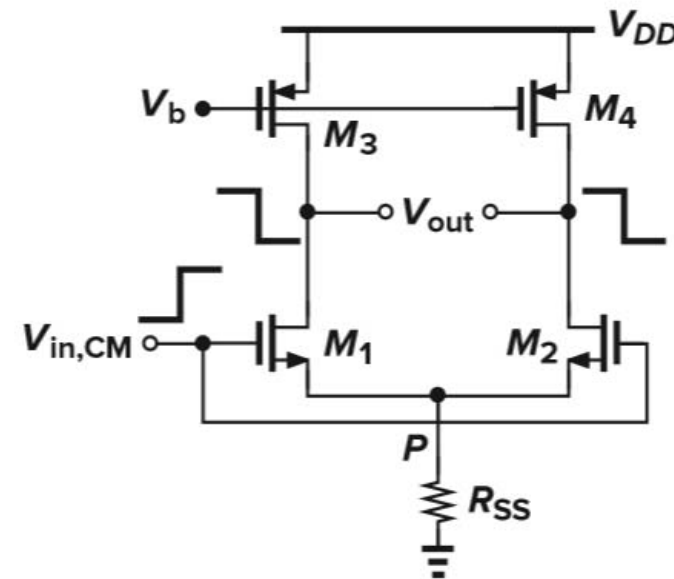
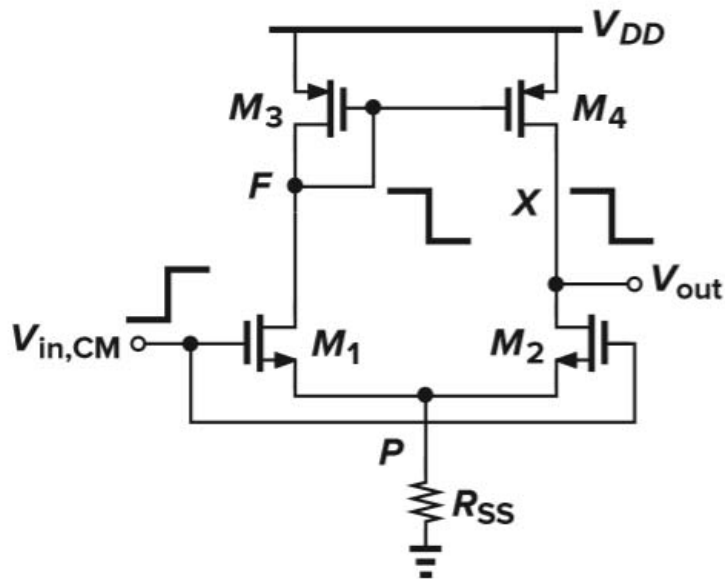
$$= g_{m3,4} (r_{o1,2} \parallel r_{o3,4}) (1 + 2g_{m1,2} R_{SS})$$

– *Study the situation of mismatches by yourself*



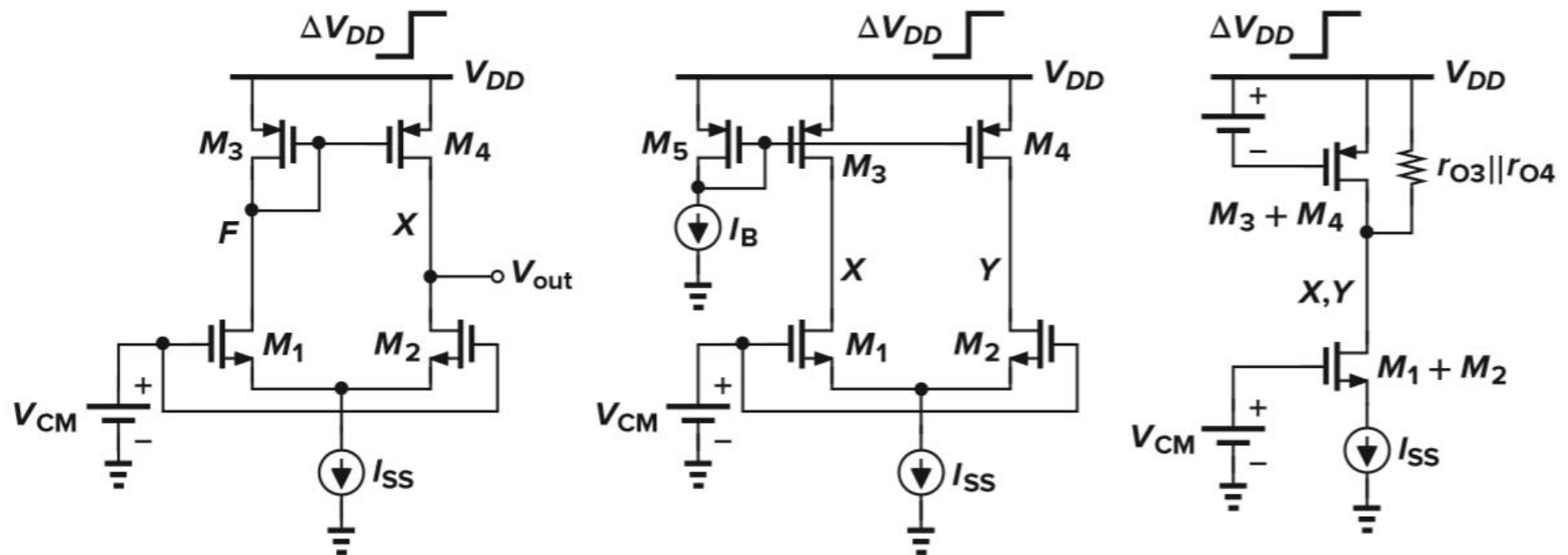
5-T AMP v.s. Fully Diff: $\Delta V_{in,CM}$

- Poor common-mode rejection than fully diff. pair



5-T AMP v.s. Fully Diff: ΔV_{DD}

- Poor supply rejection than fully diff. pair



Outline

- **Basic current mirrors**
 - **Applications**
 - **Basic structures**
 - **Drawbacks**
- **Cascode current mirrors**
 - **Basic cascode current mirrors**
 - **Low-voltage cascode current mirrors**
 - **Methods for generating the bias of LV cascode**
- **Active current mirrors**
 - **Large-signal analysis**
 - **Small-signal analysis**
 - **CM response**
- **Bias techniques**